Sentiment-augmented volatility forecasting using GARCH and Random Forests

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Chapter 1: Introduction

1.1 Background

Bitcoin has grown from being a mere technological experiment when it was first introduced in 2008/09 to a becoming a leading globally traded financial asset with significant economic relevance. Since its introduction, Bitcoin has has had phases of retail speculation, high dealers' testing, and just recently, there has been mainstream adoption of its trade even at the national levels as is the case of El Salvador where it was adopted by the state as an official currency(Argente & Van Patten, 2024). From 2017 through 2025, Bitcoin has been the market leader among cryptocurrencies and is constantly highly ranked in this class of assets as well as other top globally traded digital assets, with evolving regulatory developments as well as increasing liquidity which has resulted in enhanced adoption of Bitcoin as an asset.

Bitcoin exhibits high volatility which makes it different from typical investment assets, with the variation in daily returns frequently exceeding those observed in equities, commodities, and foreign exchange markets, with huge price swings often influenced by market sentiment, investor attention, regulatory announcements, and the broader macro-financial developments. Given that this has become such an important asset, it raises the need to understand how to accurately forecast its volatility since Bitcoin holds highly significant value for both investors and regulators alike.

1.2 Importance of volatility forecasting

Volatility forecasting is an integral part of financial economics as the accurate predictions of conditional variance support in portfolio optimisation, hedging strategies, and risk management frameworks such as Value-at-Risk (VaR). For regulators and investors alike, volatility assessments enhance a deeper understanding of speculative bubbles, systemic risks, and market stability.

Conventional econometric models, especially the Autoregressive Conditional Heteroskedasticity (ARCH) model introduced by Engle (1982) and its generalised extension, GARCH, by Bollerslev (1986), remain the foundation of volatility modeling in financial economics since these models capture core empirical regularities such as volatility clustering and fat-tailed returns. These models, however, run into obstacles brought about by the fact that cryptocurrencies often exhibit stronger persistence, asymmetry patterns distinct from equities, and non-linear, sentiment-driven fluctuations compared to the highly seasonal data points like the national accounting statistics.

1.3 Research problem and research gap

While ARCH and GARCH models have been widely applied to cryptocurrencies for example, by Dyhrberg (2016); Chu et al. (2017); Katsiampa (2017), several limitations remain. First of all, many studies rely on short or outdated datasets ending before key structural events, such as the approval of U.S. spot Bitcoin ETFs in 2024(Gensler, 2024) which attracted more informed market participants to the Bitcoin spot market, possibly leading to a change in the composition of its investors, which resulted in a record net-inflows upwards of one and a half billion US dollars(Babalos et al., 2025). Secondly, few studies directly compare GARCH-family models with machine learning methods such as Random Forests, especially in the context of sentiment-driven volatility. This clearly outlines a gap since investor behaviour has been demonstrated to influence greatly how Bitcoin performs as an asset(Katsiampa, 2017), and therefore, it necessitates the use of a more accurate variable representing sentiment, hence the role of sentiment, proxied by Google Trends and other search-based indicators, remains understudied over long time spans.

From the foregoing,therefore, arises a need for a systematic comparison of GARCH-family models and machine learning techniques, with and without sentiment variables, over an extended period of time.

1.4 Research objectives

The study has four main objectives. First, I compare the forecasting accuracy of GARCH-family models, particularly GARCH(1,1) and IGARCH(1,1) and Random Forests using Bitcoin daily data. Second, I evaluate the effects of incorporating sentiment, proxied by Google Trends, on volatility forecast accuracy, and next, I assess whether sentiment-augmented Random Forest models perform better than the traditional econometric approaches used to study volatility. Finally, I aim to provide an updated evidence on Bitcoin volatility dynamics using a comprehensive dataset encompassing major regime shifts.

1.5 The relevance of sentiment

Cryptocurrency markets, and in particular, crypto price performances, are heavily influenced by behavioural factors, media narratives, online discussions, and investor attention. Prior studies such as Mai et al. (2018) and Sapkota (2022) have shown that search intensity and social media conversations and trends can lead up to price volatility. In this sense, therefore, Google trends which show the popularity of search on Google by analysing the sample of real, anonymised searches, shows interest over time, by region, and in relation to other topics becomes a viable candidate to use as proxy for the investor attention, and is used in this study, and this integration of sentiment in the volatility models has the potential to capture otherwise unobservable market forces.

1.6 Structure of the thesis

The thesis proceeds as follows: chapter one introduces the background, motivation, research questions, and contribution. Chapter two then reviews theoretical foundations and empirical findings across econometric, behavioural, and machine-learning approaches. Chapter three describes the data, preprocessing steps, and the modeling methodology. In chapter four, I present the empirical results including descriptive statistics, model comparison, and

forecasting performance. Chapter five provides an interpretation of results and how this compares to the previous literature, implications for stakeholders, and limitations. And finally, chapter six concludes with a summary and recommendations for future work.

Chapter 2: Literature review

2.1 Introduction

Over the past four decades, volatility forecasting has been a central part of financial econometrics, owing to the need to understand, quantify, and predict time-varying uncertainty in asset returns. With the emergence of Bitcoin and other digital assets, this research area has experienced renewed attention due to the unique behavioural, structural, and technological characteristics of cryptocurrency markets which has become an attractive field of empirical research. Bitcoin, in particular, depicts extreme price swings, high-frequency sharp reversals, heavy-tailed return distributions, and heightened sensitivity to regulatory, technological, and behavioural influences, features which challenge traditional econometric models and motivate comparative evaluations involving machine learning and sentiment-enhanced approaches.

In this chapter, I provide a structured, and comprehensive review of the academic literature covering the foundational ARCH and GARCH theoretical canon, early cryptocurrency applications, asymmetry and regime dynamics, realised volatility models, behavioural and sentiment-driven forecasting, and the increasing use of machine learning and other hybrid models in the forecasting of volatility

Section 2.2 revisits the foundational ARCH and GARCH literature, outlining their theoretical principles and empirical implications. Section 2.3 discusses the first-generation cryptocurrency volatility studies that applied classical GARCH models to Bitcoin and other digital assets. Section 2.4 reviews contributions on asymmetry, smooth-transition dynamics, and realised volatility. Section 2.5 reviews behavioural, sentiment, and attention-based research. Section 2.6 examines machine learning and hybrid GARCH–ML models. Section 2.7 concludes by identifying the gaps addressed by this study.

2.2 Foundational ARCH/GARCH canon (1982–1994)

Financial return series often exhibit features inconsistent with classical homoskedastic linear models, and this necessitates the developments in the study of volatility and this inconsistency is often due to volatility clustering, heavy-tailed distributions, and serial dependence in higher-order moments. The foundational literature like the introduction of ARCH established the theoretical and empirical pillars that continue to support the contemporary financial econometric studies.

2.2.1 ARCH models

The seminal contribution of Engle (1982) introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model which marked a major shift from constant-variance assumptions that had been prominent in prior literature. In Engle's specification, the conditional variance is modeled as a linear function of lagged squared residuals, a structure that captures the volatility clustering in which periods of high volatility followed by high volatility and produces fat-tailed unconditional return distributions even under normally distributed innovations. Engle's application to U.K. inflation demonstrated that macroeconomic uncertainty portrays strong temporal dependence and can be statistically modeled rather than treated as exogenous noise.

A number of key theoretical insights from the ARCH model came to the forefront. First of all, the comparison of the conditional versus the unconditional variance which made it possible for volatility to become forecastable based on past shocks. In this regard, the ARCH processes were also shown to be leptokurtic, and with fat tails even when the shocks are Gaussian. And to check for the reliability of the ARCH model and whether it has unbiased results, Engel introduced the ARCH-LM test which still remains a standard diagnostic tool in econometrics.

In spite of the new pathways that the ARCH approach introduced for empirical finance, practical estimation still becomes cumbersome because capturing long memory requires many lags(ARCH(q) with large q), making estimation of these long memory models inefficient.

2.2.2 GARCH and IGARCH

To allow the conditional variance to depend on both the lagged squared residuals and lagged conditional variance terms, Bollerslev (1986) introduced the Generalized ARCH (GARCH) model, which addressed the practical limitations of high order ARCH. Thus, the GARCH(1,1) quickly became the primary model of volatility modeling due to its highly parsimonious and empirical effectiveness. The GARCH(1,1) model often captures long-range dependencies with only two parameters, α and β, addressing parsimony and to address volatility persistence, the model uses the sum of $\alpha+\beta$ to quantify the speed of mean reversion, and the model shows that unconditional distributions remain fat-tailed.

Subsequently, Engle & Bollerslev (1986) further extended the theory by introducing the Integrated GARCH (IGARCH) model, in which the persistence parameter $\alpha+\beta = 1$. In this specification, instead of volatility shocks decaying over time, they are allowed to accumulate indefinitely, implying that conditional variance behaves like a unit-root process. Ideally, IGARCH captures environments in which markets do not revert quickly to long-run volatility levels, and where new information, including small shocks, produces long-lasting effects on uncertainty.

The development of the IGARCH framework was particularly important for modeling exchange rates in the 1980s, where persistent volatility and slow mean reversion were empirically pervasive. In subsequent years, researchers found that cryptocurrencies exhibit remarkably similar characteristics: high persistence, weak mean reversion, and volatility that remains elevated long after major shocks. IGARCH, thus, serves as a bridge between classical GARCH methodologies and the empirical reality of digital asset markets, in which volatility dynamics often resemble near-unit-root behaviour.

2.2.3 Continuous-time links

To bridge the discrete-time GARCH with the continuous-time stochastic volatility models, Nelson (1990) introduced the exponential GARCH(EGARCH), showing that as the sampling intervals shrink, many GARCH processes converge to diffusion processes. This discovery provided the theoretical justification for using GARCH in contexts traditionally

dominated by continuous-time models, such as option pricing. Nelson equally proved that GARCH(1,1) converges to a process with an inverted-gamma stationary distribution, while exponential-GARCH processes yield log-normal variance diffusions, thus providing more insights into the theoretical legitimacy for applying GARCH to high-frequency data and enriched the versatility of volatility models.

2.2.4 Stylised facts

In the subsequent years, Bollerslev et al. (1994) defined a set of stylised facts about financial return series after surveying more than a decade of empirical research, showing that these returns remain stable across assets, markets and time periods. These findings still serve as a guide for the specification, estimation and evaluation of volatility models in the GARCH tradition and beyond.

Volatility clustering is an essential fact associated with financial asset returns as large price movements tend to be followed by large movements, either negative or positive, while periods of stability and peace tend to persist. This temporal dependence in the conditional variance tends to be one of the strongest characteristics of financial returns and forms the core empirical motivation for GARCH models as it suggests that volatility is persistent and forecastable, contradicting the notion of constant variance assumed in early asset pricing models prior to the introduction of the ARCH framework.

Bollerslev and the co-authors identified that leverage effects are integral in financial markets since in these markets, volatility responds asymmetrically to news and bad news, in particular, with negative returns typically increases volatility more than good news. The authors attribute this phenomenon to financial leverage, highlighting that a fall in equity value increases leverage and raises risk, or to behavioural factors such as asymmetric reactions to losses. In this respect, therefore, even though leverage effects are a defining feature in equities, they are notably weaker or even absent in Bitcoin and many cryptocurrencies, highlighting the fundamental differences in market microstructure.

Bollerslev and the co-authors found that empirical return distributions exhibit kurtosis far exceeding that of the normal distribution, meaning that the results are heavily fat-tailed and the financial returns display more frequent extreme movements than the Gaussian

models predict as the GARCH processes with conditional normal errors can already generate heavy-tailed unconditional distributions, a feature that is further amplified when using fat-tailed innovation distributions like the Student-t.

The financial markets are found to rely heavily on information flow, and they experience predictable spikes in volatility around scheduled events such as monetary policy announcements, labour market releases, or earnings reports. Whereas these predictable bursts do not imply that the effect of information is neither uniform nor random, they have motivated the introduction of models that explicitly incorporate exogenous information or calendar effects, features that are particularly relevant to cryptocurrencies, where regulatory announcements, exchange hacks, and protocol updates often generate disproportionate volatility.

Bollerslev et al. (1994) found the existence of volatility co-movements across assets, noting that the volatilities across different assets, sectors, or markets tend to move together, which shows the presence of common shocks or shared macroeconomic risks that necessitates the need for multivariate GARCH models such as CCC, DCC, and BEKK, and explains why systemic or contagion effects can emerge during periods of market stress. This co-movement is also observed across cryptocurrencies, which often respond simultaneously to global risk sentiment or major news events.

These stylised facts jointly form the conceptual backbone for modern volatility modeling, justifying the need for dynamic conditional variance models like GARCH, as well as more flexible nonlinear approaches such as regime-switching, smooth-transition models, realised volatility frameworks, and machine learning methods. The persistence, asymmetry, tail behaviour, and cross-asset dependence described by these facts are very present in cryptocurrency markets, making them crucial considerations in any attempt to model Bitcoin volatility.

2.3 First generation cryptocurrency volatility studies (2016–2019)

The initial research on cryptocurrencies applied the classical GARCH-family models with the aim of assessing whether Bitcoin behaved like traditional assets or not, identifying important empirical characteristics like extreme volatility, persistence patterns, and possible regime shifts which have served as defining features for the subsequent research in this area.

2.3.1 Symmetry and volatility persistence

To position the asset within the broader volatility landscape, Dyhrberg (2016) carried out one of the earliest thorough GARCH applications to Bitcoin, comparing its behaviour to gold and the U.S. dollar. Using both GARCH(1,1) and EGARCH specifications, the study highlighted several important features that have shaped much of the early cryptocurrency volatility literature.

First, Dyhrberg discovered the presence of high volatility persistence similar to commodity markets, implying that shocks to Bitcoin variance disappear slowly and can influence volatility over extended periods, a finding which has been found to be consistent with the later findings as in Chu et al. (2017) and Katsiampa (2017), which frequently identified IGARCH-type behaviour for Bitcoin and other cryptocurrencies.

Second, the existence of symmetric volatility responses, which indicate that positive and negative return shocks have similar effects on the volatility of Bitcoin, sharply contrasts with equity markets, where negative shocks typically trigger larger volatility increases. In their subsequent study, Cheikh et al. (2020) confirmed this phenomenon and even documented *inverted asymmetry* for certain cryptocurrencies, where volatility sometimes increases more after positive news, a feature contradicting the traditional leverage effect in financial markets.

Finally, since Bitcoin exhibited weak correlations with other equities, Dyhrberg (2016) suggested that it can be used potentially as a diversifying asset and hedge against equities, thus assuming roles similar to the precious metals like gold, which sets the stage for exploring how Bitcoin as an asset behaves in relation to other assets as well as macroeconomic and

behavioural variables.

From the foregoing, therefore, Dyhrberg's results imply that the absence of leverage effects hints at Bitcoin's distinctive behavioural and informational environment, characterised by retail-driven sentiment, speculative trading, and limited balance sheet mechanisms compared to equities, a fact consistent with later studies showing that sentiment and attention variables such as forum discussions as presented by Mai et al. (2018), news narratives by Sapkota (2022) and Coulter (2022), and Google search intensity play a first-order role in driving volatility.

The results also show that, the symmetry of volatility responses suggested early on that classical GARCH extensions designed for leverage effects (e.g., EGARCH, GJR-GARCH) might be less relevant for Bitcoin. Indeed, empirical comparisons in Chu et al. (2017) and Katsiampa (2017) often found that symmetric models (GARCH, IGARCH, CGARCH) performed as well or even better than asymmetric ones.

Finally, Dyhrberg's work established Bitcoin as an asset whose volatility behaviour is closer to commodities than equities, thereby reinforcing the need for models capable of capturing persistent, sentiment-sensitive, and regime-dependent dynamics which serves as the basis for later research using regime-switching GARCH models (Ardia et al., 2019), smooth-transition models (Cheikh et al., 2020), and hybrid machine learning approaches (Zahid et al., 2022; Han et al., 2025), all of which attempt to account for nonlinearities and evolving market conditions not captured by early single-regime models.

2.3.2 Comprehensive GARCH comparisons

In carrying out one of the first large-scale, systematic comparisons of the GARCH-type models applied to digital assets, Chu et al. (2017) estimated twelve volatility specifications, including GARCH, EGARCH, GJR-GARCH, FIGARCH, and IGARCH across seven major cryptocurrencies, with results revealing several distinctive features of cryptocurrency volatility dynamics. In the first place, IGARCH frequently emerged as the best model for modeling the volatilities of cryptocurrencies, suggesting that volatility shocks in Bitcoin and other cryptocurrencies exhibit near-unit-root persistence, a finding consistent with the long-memory properties of Bitcoin which was later explored through component and

regime-switching models. Secondly, normal innovations were found to often outperform heavy-tailed alternatives, a somewhat surprising result given the strong empirical evidence of fat tails in traditional equity and commodity returns. This result may reflect the dominance of structural breaks, speculative bursts, and regime shifts in crypto markets rather than heavy-tailed shock distributions per se. The different cryptocurrencies were found to exhibit different volatility structures, with some assets displaying stronger asymmetry, longer memory, or higher sensitivity to shocks than others. This heterogeneity suggests that cryptocurrencies cannot be modeled as a uniform asset class given that they may reflect differences in liquidity, maturity, and investor composition, key factors which influence the performance of the returns of cryptocurrencies.

Thus, Chu and the co-authors positioned IGARCH as a natural benchmark for Bitcoin volatility modeling and highlighted the need for models capable of capturing long persistence and structural heterogeneity. In an indirect response to this study, subsequent research including component GARCH models by Katsiampa (2017), regime-switching GARCH frameworks by Ardia et al. (2019), and smooth-transition approaches as in Cheikh et al. (2020) bring more light to the empirical complexities that Chu et al. (2017) identified.

2.3.3 Component GARCH

In comparing nine GARCH-type models including the standard GARCH, EGARCH, GJR-GARCH, TGARCH, and FIGARCH, Katsiampa (2017) found that the Component GARCH (CGARCH) model offered the best overall fit among these GARCH-family models for cryptocurrencies. CGARCH decomposes volatility into a short-run transitory component and a long-run persistent component, thereby capturing both immediate reactions to market news and deeper structural trends. The finding that Bitcoin's volatility is driven by both short-lived shocks like speculative bursts, exchange incidents, regulatory announcements and slowly evolving forces for example, adoption rates, liquidity growth, macroeconomic uncertainty provides an important conceptual bridge between earlier research work.

In particular, the CGARCH superiority aligns closely with evidence from Chu et al. (2017), who found the existence extreme persistence and near-unit-root behaviour in cryptocurrencies, often favouring IGARCH specifications. Katsiampa's results refine this

view by showing that what appears to be IGARCH-type persistence in single-regime models may actually reflect a combination of long-term variance dynamics and short-term speculative noise. This is a consistent interpretation as was the case in Dyhrberg (2016), who also observed persistent volatility and symmetric responses to shocks, suggesting that Bitcoin's volatility is not purely driven by episodic events but reflects deeper market forces.

Thus, Katsiampa's contribution serves as an intermediate step between the early single-regime GARCH evidence and the nonlinear, regime-switching models that followed in later studies.

2.3.4 Regime switching

Ardia et al. (2019) provided evidence that Bitcoin's volatility dynamics are regime-dependent by identifying several low-volatility and high-volatility regimes, highlighting through Markov-switching GARCH models (MSGARCH) that regime switching explains the appearance of IGARCH-like persistence in single-regime models with high-volatility regimes coinciding with market crises and speculative booms. The authors also show that Value-at-Risk forecasts improve under regime-switching models, findings which have motivated later research on nonlinear models and hybrid specifications.

2.4 Asymmetry, smooth transitions, and realised volatility (2019–2022)

Volatility in cryptocurrencies is often asymmetric, persistent, and subject to structural breaks, hence the literature has expanded to include models with nonlinear and smooth-transition dynamics, as well as realised volatility models that exploit high-frequency data.

2.4.1 Asymmetric and smooth-transition dynamics

Cheikh et al. (2020) introduced Smooth Transition GARCH (ST-GARCH) models to analyse gradual, rather than abrupt, shifts between low-volatility and high-volatility

states in major cryptocurrencies and found that in the first place, Bitcoin, Ripple and Litecoin portray inverted asymmetry where volatility increases more after positive returns, contrasting strongly with the negative-return leverage effects observed in equities. They also found that volatility responses can vary across cryptocurrencies depending on maturity, liquidity and use base as depicted by Ethereum that exhibits negligible asymmetry and that Bitcoin also exhibited smoother transitions between volatility states than smaller cryptocurrencies which indicates a relatively more stable and liquid market structure.

This is a confirmation of the existence of symmetry in Dyhrberg (2016) and the dominance of symmetric GARCH models in Chu et al. (2017), while extending the literature by showing that asymmetry in crypto markets is not only weak but, in some cases, reversed. The inverted asymmetry pattern supports interpretations of speculation, heavy reliance on momentum-driven trading behaviour, and reinforces the view that volatility in cryptocurrencies is shaped more by behavioural dynamics than by conventional leverage mechanisms.

2.4.2 Realised volatility models

The realised volatility models gained prominence in cryptocurrency research due to their ability to capture intraday information. Corsi (2009) proposed the Heterogeneous Autoregressive (HAR-RV) model, which approximates long-memory volatility using daily, weekly, and monthly realised volatilities and the subsequent studies applying HAR-RV to Bitcoin found that HAR-RV captures long-memory effects without fractional integration and often, rivals or outperforms GARCH models in out-of-sample forecasts, concluding that high-frequency realised volatility reduces noise and improves forecast accuracy. Later, Dudek et al. (2024) would show further proof that hybrid HAR-ML variants have more improvements on the standalone HAR-RV models.

2.5 Behavioural and sentiment drivers (2018–2025)

Since the cryptocurrency market is heavily influenced by online narratives, media cycles, social media activities and investor attention, behavioural finance plays an integral role in

pricing and valuation of cryptocurrencies. This section, therefore, covers research on forums, news, Google trends, topic modeling, and sentiment–econometric hybrids.

2.5.1 Social media and forums

Mai et al. (2018) analysed BitcoinTalk forum posts and found that the sentiment from less active ("silent majority") users strongly predicts next-day prices, implying that there are "key" investors who directly influence the price of Bitcoin simply by what they say or post online. On the other hand, highly active users or the "vocal minority" influence trading volume, the actual raw numbers of Bitcoin units traded per day, and not the returns. The authors also show that forum sentiment provides stronger predictive power than Twitter sentiment, a sensible finding given that the '*Bitcointalk*' that they study covers, almost exclusively, news on Bitcoin, unlike Twitter/X on which all sorts of items or news articles are shared.

Thus, Mia and the co-authors highlight the decentralised, retail-driven nature of cryptocurrency markets, where specific community opinions rather than large, influential individuals shape short-term price dynamics. This aligns with later evidence showing that crowd sentiment, search intensity, and collective attention often precede volatility spikes (Wang et al., 2023), providing a behavioural rationale for incorporating sentiment variables into volatility models by distinguishing between market participation intensity and predictive sentiments.

2.5.2 News sentiment

By extracting news articles from major English newspapers, Sapkota (2022) found that mixed positive-negative sentiment has stronger effects on the direction of prices than purely positive or negative sentiment. This is because the mixed positive-negative is a clearer reflection of the real life trading situation. The author also found that sentiment, in general, significantly improves the HAR-RV forecasts, and weekly news is more useful than the daily news in the HAR-RV forecasting framework.

The emphasis on mixed-tone narratives suggests that ambiguity and conflicting news

signals generate heightened uncertainty and speculative reactions among investors and the stronger predictive power at the weekly horizon indicates that sentiment diffusion in crypto markets may operate more slowly than in equity markets, possibly due to heterogeneous investor bases and global time-zone effects.

2.5.3 Search-based attention

Search-based measures capture the attention of retail investors and Hoang & Vo (2024) identified that abnormal search volume (ASVI) is integral in predicting next-week crypto-returns, and also influences the increase in volatility and trading volume. ASVI is also shown to have stronger effects on larger cryptocurrencies in comparison to the smaller units.

This reinforces the attention-based asset pricing literature and justifies the use of Google Trends as a timely behavioural proxy, especially in markets lacking traditional fundamental anchors. The finding that ASVI influences volatility is consistent with the "information-overload" hypothesis, where surges in retail attention precede market disorder and speculative trading. Later ML studies by Wang et al. (2023) incorporate Google Trends as an input feature, showing that attention-based variables enhance nonlinear models more effectively than standard GARCH specifications.

2.5.4 Topic modeling and narrative effects

Coulter (2022) applied Latent Dirichlet Allocation (LDA) to global news coverage and identified as dominant narrative categories crypto-crime, crypto-governance, and crypto-economy. Crypto-crime and governance narratives often preceded sharp daily declines in Bitcoin prices, especially during periods of regulatory uncertainty and exchange hacks, showing that volatility is not only driven by numerical sentiment indices but also by the thematic content and narrative framing of news, a finding consistent with Sapkota (2022) who indicated that sentiment-driven volatility depends not only on tone but also on the type of narrative being amplified.

2.5.5 Hybrid sentiment – econometric models

Hybrid sentiment–econometric models integrate behavioural indicators with traditional time-series structures. For instance, Kabo et al. (2025) showed that sentiment and GDP improve ARIMA and LSTM forecasts, highlighting the complementary value of macroeconomic and behavioural factors and Leber & Egyed (2025) demonstrated that negative sentiment Granger-causes volatility and improves VaR forecasts when incorporated into GARCH–LSTM hybrids.

These studies emphasise that sentiment affects both central tendency and tail risk. They also show that hybrid models can extract nonlinear behavioural effects that GARCH models alone cannot capture, reinforcing the motivation for this study to compare traditional econometric and machine learning approaches in a sentiment-augmented framework.

2.5.6 Random forests

Basher & Sadorsky (2022) found that Random Forest(RF) models outperform logistic regression in predicting Bitcoin price direction and this is, in part, because Random Forest models handle nonlinear interactions and high-dimensional data and are robust to multicollinearity and noise, hence provide interpretable feature importance measures.

RF has also been used to model realised volatility by Luong & Dokuchaev (2018) and forecast volatility in emerging markets by Diane & Brijlal (2024), applications which indicate that tree-based models can successfully capture irregular patterns, structural breaks, and nonlinearities which features prevalent in Bitcoin markets, and therefore, the flexibility of RF models makes them strong candidates for sentiment-augmented volatility forecasting.

2.5.7 Deep learning

Deep learning models such as LSTM, GRU, and CNN architectures outperform traditional econometric models in many studies. Kim & Won (2018) proposed LSTM–GARCH hybrids and subsequently, Wang et al. (2023) found that LSTM models outperform GARCH for weekly and monthly volatility. Eventually, in an attempt to improve accuracy, Han et al.

(2025) used GARCH forecasts as LSTM inputs, and the result was a drastic improvement in the prediction accuracy of the LSTM model.

These improvements arise because deep learning architectures capture long-memory patterns, nonlinear relationships, and structural regime changes which are dynamics often persistent in Bitcoin volatility and amplified by sentiment shocks.

2.5.8 Hybrid GARCH – machine learning(ML) models

In an attempt to improve the forecasting powers of the traditional econometric models, researchers have proposed hybrid models, combining the structural interpretability of these econometric models with the adaptability of the machine learning approaches. By assessing the GARCH residual learning, Zahid et al. (2022) showed that ML models learn residual patterns not captured by GARCH. And later on, to incorporate news sentiments and their effects on the model performance, Leber & Egyed (2025) introduced the sentiment-augmented hybrids and reported that these behavioural variables improve performance.

2.5.9 Random forests for volatility

Random Forests(RF) have been used not only for classification but also for volatility regression and both Amoanu (2025) and Demirer et al. (2022) show that RF models can outperform econometric models for certain financial time series, especially when nonlinear predictors or behavioural indices are included. In realised volatility forecasting, RF improves over HAR-JL specifications by capturing nonlinear interactions (Luong & Dokuchaev, 2018), thus reinforcing the potential value of RF in modeling Bitcoin volatility, particularly when sentiment variables are incorporated.

More flesh here? Seems too brief!

2.6 Research gap

The above mentioned literature reviewed reveals that Bitcoin volatility is extraordinarily persistent, often approximating IGARCH behaviour and that volatility dynamics are nonlinear, asymmetric, and regime-dependent. The literature also shows that sentiment which is often captured in the reported literature through search activity, news tone, and forums, is a major driver of volatility and that machine learning and in particular, hybrid models, consistently outperform standalone econometric approaches.

Yet, despite these clear indications, important gaps still remain in literature. Few studies directly compare GARCH-family models with Random Forests for volatility forecasting and sentiment is underutulised or rarely integrated systematically across model classes for longer datasets spanning major post-2023 structural events. Therefore, this thesis addresses these gaps by conducting a sentiment-driven comparative analysis of GARCH and Random Forest models using Bitcoin data from 2017–2025.

Chapter 3: Methodology

3.1 Introduction

This chapter presents the econometric and machine learning methodology used to forecast Bitcoin volatility and to evaluate whether sentiment-enhanced nonlinear methods outperform standard GARCH benchmarks. The chapter specifies the data-generating process for returns, the conditional variance dynamics under GARCH-type models, the Random Forest forecasting set-up, the out-of-sample forecasting design, and the forecast evaluation framework and multiple-model comparison procedures.

3.2 Notation and return construction

Let P_t denote the daily closing price of Bitcoin on day t. The continuously compounded (log) return is defined as

$$r_t = 100 \left(\ln P_t - \ln P_{t-1} \right), \tag{3.1}$$

where the factor 100 expresses returns in percentage terms.

Let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. This specification follows standard practice in financial econometrics, where returns are modeled as conditionally heteroskedastic processes with time-varying variance as in Bollerslev (1986).

A general conditional mean specification is

$$r_t = \mu + \sum_{i=1}^{p} \phi_i r_{t-i} + \varepsilon_t, \tag{3.2}$$

where ε_t is the innovation satisfying

$$\mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0, \qquad \mathbb{V}(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2. \tag{3.3}$$

The focus of the study is on modeling and forecasting σ_t^2.

3.3 Realised volatility proxy

Since true volatility is latent, evaluation is based on a realised volatility proxy constructed from returns. A standard daily proxy is the squared return,

$$RV_t = r_t^2, \tag{3.4}$$

In robustness analysis, alternative proxies based on rolling windows are considered, for example

$$RV_t^{(h)} = \frac{1}{h} \sum_{j=0}^{h-1} r_{t-j}^2, \tag{3.5}$$

for horizon $h \in \{5, 10, 20\}$. Although squared returns are a noisy proxy for latent volatility, they are widely used in daily-frequency studies where intraday data are unavailable (Patton, 2011).

3.4 Econometric volatility models

3.4.1 Shock decomposition and innovation distribution

Innovations are written as

$$\varepsilon_t = \sigma_t z_t, \tag{3.6}$$

where $\{z_t\}$ is i.i.d. with $\mathbb{E}(z_t) = 0$ and $\mathbb{V}(z_t) = 1$. To accommodate fat tails commonly observed in cryptocurrency returns, estimation is conducted under alternative distributional assumptions for z_t, including the Gaussian and Student-t distributions:

$$z_t \sim \mathcal{N}(0, 1) \quad \text{or} \quad z_t \sim t_\nu \text{ scaled to unit variance.} \tag{3.7}$$

Allowing for Student-t innovations accommodates the excess kurtosis commonly observed in cryptocurrency returns and is standard in volatility modeling (B. E. Hansen, 1994).

3.4.2 GARCH(1,1)

The baseline conditional variance specification is GARCH(1,1):

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{3.8}$$

with parameter restrictions

$$\omega > 0, \qquad \alpha \geq 0, \qquad \beta \geq 0. \tag{3.9}$$

A sufficient condition for covariance stationarity is

$$\alpha + \beta < 1, \tag{3.10}$$

in which case the unconditional variance exists and equals

$$\mathbb{E}(\sigma_t^2) = \frac{\omega}{1 - \alpha - \beta}. \tag{3.11}$$

The persistence of volatility shocks is measured by $\alpha+\beta$, and the corresponding half-life can be reported as

$$HL = \frac{\ln(0.5)}{\ln(\alpha + \beta)}. \tag{3.12}$$

3.4.3 IGARCH(1,1)

To allow near-unit-root volatility behaviour, the IGARCH(1,1) model imposes

$$\alpha + \beta = 1, \tag{3.13}$$

so that

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + (1 - \alpha)\sigma_{t-1}^2. \tag{3.14}$$

Under IGARCH, shocks to volatility decay very slowly, which may reflect the high persistence documented in Bitcoin volatility and since near-unit-root volatility behaviour has been

documented in cryptocurrency markets, it motivates the use of IGARCH-type specifications to capture highly persistent volatility dynamics.

3.4.4 Sentiment-augmented variance dynamics

To test whether investor attention contains incremental information for volatility dynamics, sentiment is incorporated via an exogenous regressor S_{t-1}. A parsimonious sentiment-augmented specification is

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \gamma S_{t-1}, \tag{3.15}$$

where γ captures the marginal contribution of attention shocks to next-day conditional variance. Nonlinearities can be partially addressed by including transformations such as ΔS_{t-1} or $\ln(1 + S_{t-1})$.

3.5 Estimation via maximum likelihood

Let θ denote the vector of parameters (including ω, α, β and distributional parameters such as ν for Student-t). Conditional on $\mathcal{F}_{t-1}$, the log-likelihood contribution is

$$\ell_t(\theta) = \log f(z_t(\theta)) - \frac{1}{2}\log(\sigma_t^2(\theta)), \qquad z_t(\theta) = \frac{\varepsilon_t}{\sigma_t(\theta)}, \tag{3.16}$$

where $f(\cdot)$ is the density of z_t. The sample log-likelihood is

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \ell_t(\theta), \tag{3.17}$$

and $\widehat{\theta}$ is obtained by maximising $\mathcal{L}(\theta)$ subject to parameter restrictions. Maximum likelihood estimation is used since it is standard for GARCH-type models and yields consistent and asymptotically normal estimators under regularity conditions (Bollerslev, 1986).

3.6 Forecast definition

A one-step-ahead conditional variance forecast from an estimated GARCH-type model is

$$\widehat{\sigma}^2_{t+1|t} = \widehat{\omega} + \widehat{\alpha}\widehat{\varepsilon}^2_t + \widehat{\beta}\widehat{\sigma}^2_t \quad \text{(plus } \widehat{\gamma} S_t \text{ if sentiment is included).} \tag{3.18}$$

These forecasts are generated recursively in a rolling-window design to mimic real-time updating.

3.7 Random Forest forecasting set-up

Let $\mathbf{X}_t$ denote the feature vector containing lagged returns, volumes, rolling volatility measures, and sentiment lags. The Random Forest produces a forecast

$$\widehat{RV}_{t+1|t} = f(\mathbf{X}_t), \tag{3.19}$$

where $f(\cdot)$ is estimated as an ensemble of regression trees. To align targets across frameworks, machine learning outputs are mapped to a variance forecast and compared to realised volatility proxies. Hyperparameters are then selected using time-series cross-validation.

3.8 Forecast evaluation: loss functions and model comparison

Forecast evaluation follows the framework proposed by Patton (2011), which emphasises the use of loss functions that are robust to latent volatility and forecast dependence. In particular, the QLIKE loss function is employed due to its desirable theoretical properties when volatility is unobserved and must be proxied.

The model comparisons are conducted using the model confidence set procedure of P. R. Hansen et al. (2011) and the superior predictive ability test of P. R. Hansen (2005), which jointly account for serial correlation, multiple testing, and data-snooping concerns.

3.8.1 QLIKE loss

Following Patton-style volatility evaluation, the primary loss function is QLIKE:

$$L_{t+1}(m) = \frac{RV_{t+1}}{\widehat{\sigma}^2_{t+1|t}(m)} - \ln\left(\frac{RV_{t+1}}{\widehat{\sigma}^2_{t+1|t}(m)}\right) - 1, \tag{3.20}$$

where $\widehat{\sigma}^2_{t+1|t}(m)$ denotes the forecast from model m.

3.8.2 Loss differentials and benchmark comparison

Let model 0 denote the benchmark (e.g., GARCH(1,1)). Define the loss differential for model m relative to the benchmark as

$$d_{t+1}(m) = L_{t+1}(0) - L_{t+1}(m). \tag{3.21}$$

A positive mean of $d_{t+1}(m)$ indicates that model m improves upon the benchmark on average. The study will use this as an additional measure of model comparison for all the ten models studied.

3.8.3 Superior predictive ability (SPA) test

The SPA test evaluates whether any competing model outperforms the benchmark while controlling for data-snooping. The null hypothesis is

$$H_0 : \max_{m \in \mathcal{M}} \mathbb{E}\left[d_t(m)\right] \leq 0, \tag{3.22}$$

meaning that all the models are worse than the benchmark model, against the alternative that at least one model has strictly positive expected improvement. The inference in this case is conducted using a block bootstrap to account for serial dependence in the loss differentials.

3.8.4 Model confidence set (MCS)

To compare multiple models simultaneously, the model confidence set procedure is used to identify a subset $\widehat{\mathcal{M}}_{1-\alpha}$ of models that are statistically indistinguishable from the best model under the chosen loss function. The MCS algorithm iteratively removes inferior models based on tests of equal predictive ability, using bootstrap-based critical values to accommodate dependence.

Chapter 4: Empirical Results

4.1 Preliminary data analysis

4.1.1 Core distributional statistics

Statistic	Returns	Realised Volatility (RV)	Lagged Sentiment
Mean	0.1515	13.5990	37.56
Median	0.1275	2.3597	36
Std. Dev.	3.6852	49.9664	18.42
Min	-46.4730	0.0000	5
Max	22.5119	2159.7414	100
Skewness	-0.6597	26.9230	0.6589
Excess Kurtosis	11.6638	1093.4381	0.3205

Table 4.1: *Core descriptive statistics for Bitcoin returns, realised volatility (proxied by squared returns), and lagged sentiment measured using Google Trends search interest.*

Presented in table 4.1 are the summary statistics for Bitcoin log returns, squared returns used as a proxy for realised volatility, and lagged Google Trends search interest which serves as the sentiment variable in the model.

The mean daily Bitcoin return over the sample period is 0.1515%, which indicates a modest positive average growth between 2017 and 2025, the standard deviation of returns being 3.6852, which highlights the substantial variability that characterises Bitcoin price movements. Similarly, the volatility proxy, measured as squared returns, displays a considerably larger mean of 13.5990 and a standard deviation of 49.9664, which also shows the extreme variability of the cryptocurrency markets and the presence of volatility clustering. The sentiment variable has an average value of 37.56 on the 0–100 scale with a standard deviation of 18.42, which suggests moderate fluctuations in investor attention over the sample period.

The higher-order moments reveal pronounced deviations from normality in the return

series, with the returns having negative skewness (-0.6597), which shows that large negative shocks occur slightly more frequently than positive ones and the excess kurtosis of 11.6638 indicates extremely fat tails relative to the normal distribution, implying that extreme price movements occur much more frequently than would be predicted by a Gaussian process.

In addition to these, the volatility proxy displays even more extreme distributional features with extremely high right-skewness (26.9230) and extraordinarily large excess kurtosis (1093.4381), which imply the presence of occasional but very large volatility spikes.

The sentiment variable, in contrast, exhibits comparatively moderate skewness (0.6589) and low excess kurtosis (0.3205), which shows that unlike the return series, it has a much smoother distribution and since Google Trends data are bounded between zero and one hundred, the sentiment series does not display the explosive tail behaviour observed in the volatility proxy. This difference suggests that the relationship between sentiment and volatility may operate through indirect or nonlinear mechanisms rather than simple linear co-movement.

4.1.2 Bitcoin price dynamics

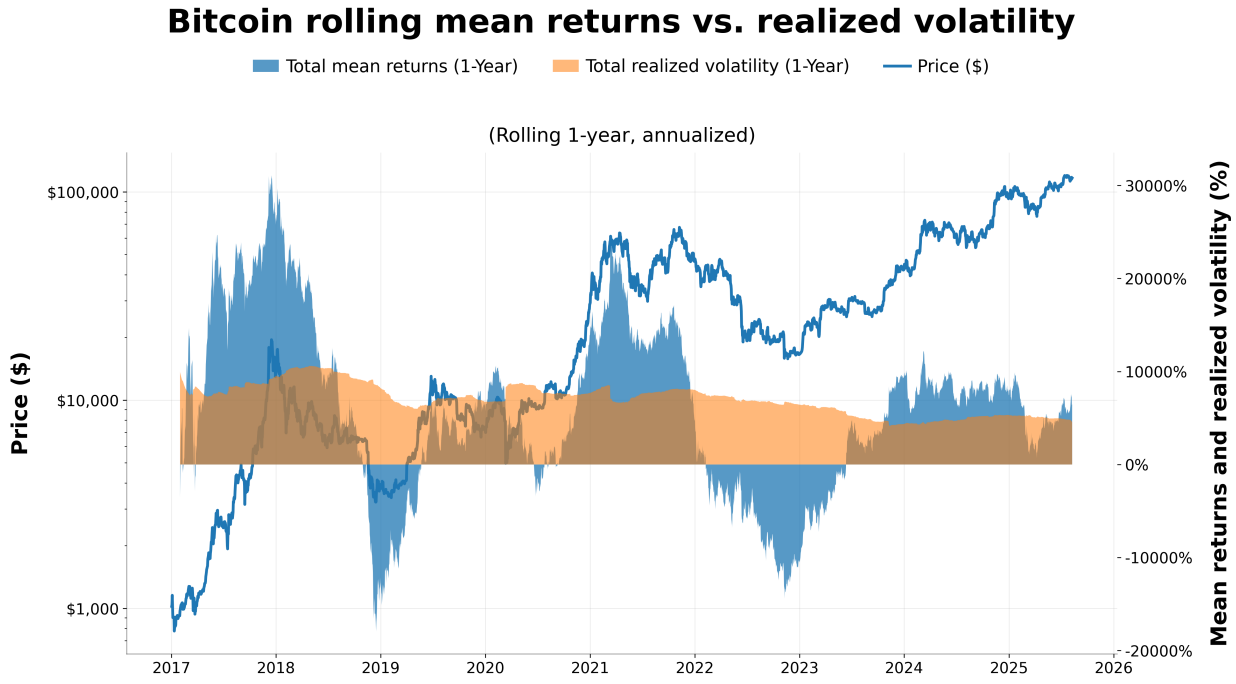


Figure 4.1: *Bitcoin price dynamics, rolling mean returns and realised volatility*

Figure 4.1 illustrates the relationship between Bitcoin price dynamics, rolling mean returns, and realised volatility using one-year rolling windows, combining three elements: the Bitcoin price series (left axis, logarithmic scale), the annualised rolling mean returns, and the annualised realised volatility computed from squared daily returns (right axis). It reveals that periods of strong price appreciation coincide with elevated average returns and increased volatility, a pattern that is particularly visible during the major bull market cycles around 2017–2018 and 2020–2021, where large positive rolling returns are accompanied by pronounced spikes in realised volatility, reflecting the speculative nature of these episodes, periods during which rapid price increases are associated with heightened uncertainty and large price fluctuations.

The figure also shows that volatility exhibits clear clustering over time as periods of high volatility tend to persist for extended intervals rather than fluctuate randomly before gradually declining. These clusters correspond closely with major market events and regime shifts in the Bitcoin market, such as the boom–bust cycles observed in 2018 and the subsequent recovery in 2020–2021.

Finally, the series shows a substantial variation in the magnitude of rolling mean returns across market phases. During bullish periods, the annualised rolling returns become strongly positive as evident between 2018 - 2019 and 2020 - 2021, while prolonged bearish phases generate negative rolling returns as in 2019 and 2022 - 2023. In contrast, the realised volatility remains strictly positive and typically declines during periods of relative market stability.

4.1.3 Quantiles and tail behaviour

The median daily return reported in table 4.2 is approximately 0.1275%, which is close to zero and consistent with the near-zero unconditional mean which is typically observed in financial return series. However, the tails show mild asymmetry with the 1% quantile of returns being -10.6030%, slightly exceeding the magnitude of the 99% quantile of 10.1557%, suggesting a somewhat greater downside risk, which aligns with the negative skewness reported in the descriptive statistics in table 4.1.

The volatility dynamics become substantially more pronounced when examining squared

Quantile	Returns	Realised Volatility (RV)	Lagged Sentiment
1%	-10.6030	0.0007	7
5%	-5.8732	0.0128	12
25%	-1.3578	0.3872	22
50%	0.1275	2.3597	36
75%	1.7339	10.7193	49
95%	5.8442	59.4999	70
99%	10.1557	146.3326	93

Table 4.2: *Selected quantiles illustrating the tail behaviour of Bitcoin returns, realised volatility (proxied by squared returns), and lagged sentiment.*

returns given that while the median squared return is relatively small at 2.3597, the upper tail expands rapidly as the 95% quantile reaches 59.4999 and the 99% quantile increases sharply to 146.3326, indicating that volatility escalates dramatically during extreme market episodes. On the other hand, the lower tail remains close to zero, with a 1% quantile of 0.0007, which depicts prolonged periods of relatively calm market conditions punctuated by rare but intense volatility spikes.

The quantile range for the sentiment variable lies between 7 at the 1% level and 93 at the 99% level, with a median value of 36, which is a wide dispersion indicating that investor attention fluctuates substantially over time. However, unlike the volatility proxy, the sentiment distribution does not exhibit explosive tail behaviour, owing to the 0–100 scaling inherent in Google Trends data.

The quantile evidence, therefore, reinforces the presence of heavy-tailed behaviour in Bitcoin returns and especially in volatility measures and while typical market fluctuations are captured within the interquartile range (0.3872 to 10.7193 for squared returns), the rapid escalation between the 95% and 99% quantiles highlights the severity of extreme volatility episodes that characterise cryptocurrency markets.

4.2 Dependence and persistence diagnostics

The autocorrelation function of returns in figure 4.2 decays rapidly and remains largely within the confidence bands after the first few lags, which shows that the serial dependence

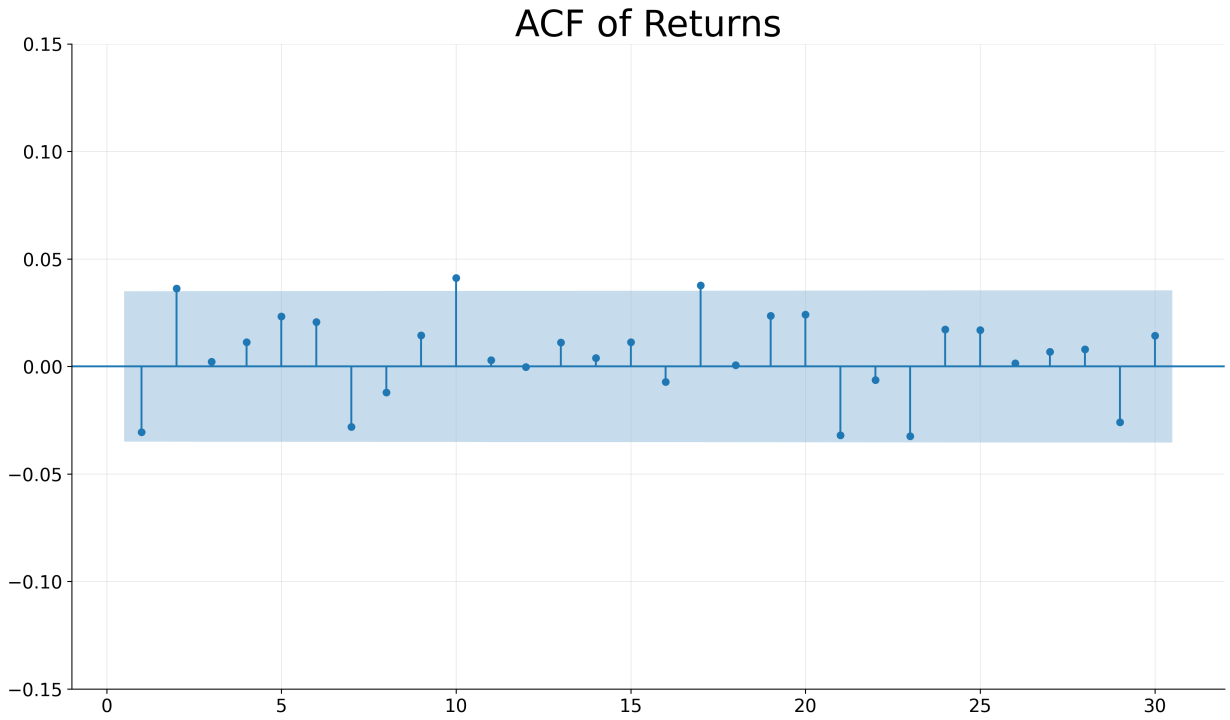


Figure 4.2: *ACF of returns*

in the conditional mean is weak and short-lived and this is supported by the Ljung–Box test results in table 4.3 where the p-values for returns remain relatively high across lag horizons (0.0344 at lag 10, 0.0948 at lag 20, and 0.0981 at lag 30), suggesting that serial dependence in raw returns is limited and becomes statistically insignificant at longer horizons, a behaviour which shows weak-form market efficiency, where past returns provide little linear predictive power for future returns. In contrast, the autocorrelation function of squared returns

	Returns	Realised Volatility (Squared Returns)
Lag 10	0.0344	0.0000
Lag 20	0.0948	0.0000
Lag 30	0.0981	0.0000

Table 4.3: *Ljung–Box test p-values for dependence diagnostics of returns and realised volatility proxied by squared returns.*

in figure 4.3 exhibits strong and persistent positive autocorrelation. In this case, the first lag shows a large correlation, followed by a gradual and slow decay that remains positive across many lags, a pattern that shows pronounced volatility clustering, whereby periods of

high volatility tend to be followed by further periods of high volatility and this is strongly confirmed by the Ljung-Box test results in table 4.3 where the p-values for squared returns are effectively zero across all reported lag horizons, leading to a clear rejection of the null hypothesis of no serial dependence.

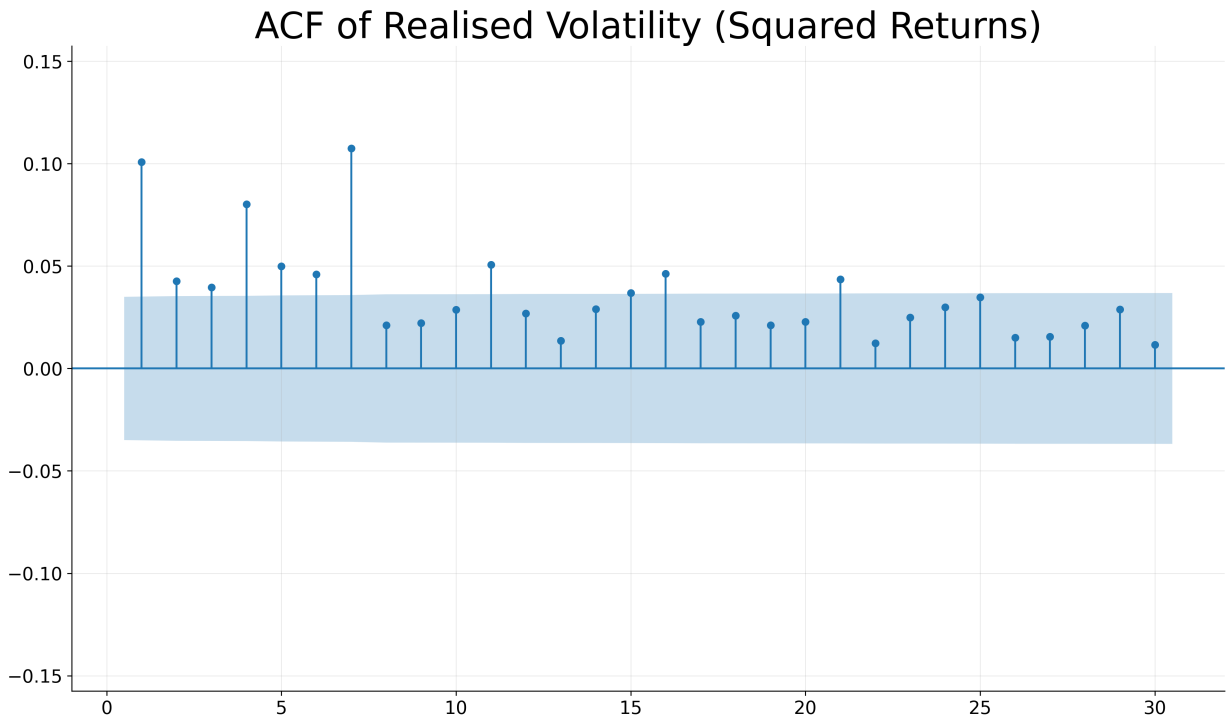


Figure 4.3: *ACF of realised volatility*

These tests reveal that while returns themselves display minimal autocorrelation, the volatility proxy exhibits strong persistence over time which provides a direct empirical justification for the use of ARCH-type models designed to capture time-varying conditional heteroskedasticity. Furthermore, the slow decay in the autocorrelation of squared returns suggests highly persistent variance dynamics, motivating the comparison between stationary GARCH(1,1) and near-integrated IGARCH specifications in the subsequent modeling section.

4.3 Distrinutional and stationarity diagnostics

4.3.1 Normality tests

Statistic	Returns	Realised Volatility (RV)
Observations	3142	3142
JB Statistic	17975.6826	156405979.3164
p-value	0.0000	0.0000

Table 4.4: *Jarque–Bera normality test results for Bitcoin returns and realised volatility (proxied by squared returns).*

The Jarque–Bera (JB) statistics reported in table 4.4 strongly reject the null hypothesis of normally distributed data for both returns and the realised volatility proxy with a JB statistic of 17 975.68 for the returns with a p-value of 0.0000, indicating significant departures from normality and the deviation is even more pronounced for squared returns, where the JB statistic exceeds 156 million, reflecting extremely high skewness and kurtosis. These further confirm the heavy-tailed nature of Bitcoin return dynamics and the extreme right-skewness present in volatility proxies which imply that extreme observations are more frequent in occurance than would be predicted under a standard Gaussian distribution. Consequently, assuming normally distributed innovations would underestimate the probability of large market movements which motivates the use of heavy-tailed innovation distributions, such as the Student-t, in the estimation of GARCH-type models.

While the Jarque–Bera results clearly indicate strong departures from normality in the unconditional distribution of returns and volatility proxies, this does not imply that heavy-tailed innovation distributions will necessarily improve forecasting performance in GARCH-type models as the JB test captures unconditional properties of the data, whereas GARCH models operate in a conditional framework. Once volatility clustering is accounted for, much of the excess kurtosis is absorbed by the variance dynamics, meaning that the standardized residuals need not exhibit the same degree of non-normality as the raw series.

Moreover, although Student-t innovations improve in-sample fit, as reflected in higher likelihood values and lower information criteria, this additional flexibility can come at the

cost of poorer out-of-sample performance.

4.3.2 Stationarity diagnostics

Statistic	Returns	Realised Volatility (RV)	Lagged Sentiment
ADF Statistic	-38.8121	-16.7408	-4.6501
p-value	0.0000	0.0000	0.0001
Used Lags	1	6	29
Observations	3140	3135	3112
Critical Value (5%)	-2.8625	-2.8625	-2.8625

Table 4.5: *Augmented Dickey–Fuller (ADF) unit root test results for returns, realised volatility (proxied by squared returns), and lagged sentiment.*

The ADF test results in table 4.5 reject the null hypothesis of a unit root for all variables at conventional significance levels where the returns display particularly strong evidence of stationarity with an ADF statistic of -38.8121 and a p-value of 0.0000. Similarly, the volatility proxy and the sentiment variable also reject the unit root null with ADF statistics of -16.7408 and -4.6501 respectively and p-values well below 0.05 in both cases. This indicates that all series are stationary in levels and do not require differencing before modeling.

Statistic	Returns	Realised Volatility (RV)	Lagged Sentiment
Observations	3142	3142	3142
KPSS Statistic	0.1430	2.1518	1.2838
p-value	0.1000	0.0100	0.0100
Used Lags	8	17	31

Table 4.6: *KPSS stationarity test results for Bitcoin returns, realised volatility (proxied by squared returns), and lagged sentiment.*

To provide a complementary assessment of stationarity, the KPSS test results reported in table 4.6 show that for the returns series, the KPSS statistic is 0.1430 with a p-value of 0.1000, indicating that the null hypothesis of stationarity cannot be rejected while the KPSS statistics for realised volatility (2.1518) and lagged sentiment (1.2838) are associated with p-values of 0.0100, leading to rejection of the stationarity null for these variables.

The divergence between the ADF and KPSS results suggests that while volatility and sentiment do not contain unit roots, they may exhibit structural shifts or evolving regimes over the sample period which may further prove the presence of volatility clustering and changing market conditions in cryptocurrency markets.

4.3.3 Correlation analysis

Variable	Returns	Absolute Returns	Realised Volatility (RV)	Lagged Sentiment
Returns	1.0000	-0.0326	-0.1569	-0.0197
Absolute Returns	-0.0326	1.0000	0.7690	0.0213
Realised Volatility (RV)	-0.1569	0.7690	1.0000	0.0040
Lagged Sentiment	-0.0197	0.0213	0.0040	1.0000

Table 4.7: *Pearson correlation matrix for Bitcoin returns, absolute returns, realised volatility (proxied by squared returns), and lagged sentiment.*

As shown in table 4.7, returns display a modest negative correlation with realised volatility (-0.1569), meaning that negative price movements tend to coincide with higher volatility, while the absolute returns and squared returns, on the other hand, exhibit a strong positive correlation (0.7690), which confirms that both variables capture closely related information about the magnitude of price fluctuations and both measures behave in a similar manner as volatility proxies, although squared returns amplify extreme movements relative to absolute returns.

The sentiment variable shows only very weak contemporaneous correlations with the financial variables, with a negative correlation with the returns(-0.0197), while correlations with absolute returns (0.0213) and realised volatility (0.0040) are close to zero, meaning that Google trends as a sentiment variable does not exhibit a strong linear relationship with Bitcoin price movements or volatility.

To provide a non-parametric measure of association that is less sensitive to extreme observations, table 4.8 presents the Spearman rank correlation matrix, which displays broadly similar patterns to the Pearson correlations in table 4.7. The return series maintains weak correlations with volatility measures, while absolute returns and squared

Variable	Returns	Absolute Returns	Realised Volatility (RV)	Lagged Sentiment
Returns	1.0000	0.0754	0.0754	-0.0335
Absolute Returns	0.0754	1.0000	1.0000	0.0030
Realised Volatility (RV)	0.0754	1.0000	1.0000	0.0030
Lagged Sentiment	-0.0335	0.0030	0.0030	1.0000

Table 4.8: *Spearman rank correlation matrix for Bitcoin returns, absolute returns, realised volatility (proxied by squared returns), and lagged sentiment.*

returns remain perfectly aligned due to their closely related construction. On the other hand, sentiment variable again shows negligible correlations with the financial variables, with coefficients close to zero and a negative correlation with the return series(-0.0335).

The similarity between the Pearson and Spearman results indicates that the weak relationships observed are not primarily driven by extreme outliers, but instead, they suggest that sentiment does not exhibit a strong contemporaneous monotonic relationship with Bitcoin volatility or returns, implying that any predictive influence of sentiment is more likely to be through non-linear relationships or lag structures.

4.4 Model estimation results

4.4.1 Model specification and estimation setup

Table 4.9 summarises the empirical modeling framework adopted in this study where log returns are used as the primary return variable, while squared returns serve as the proxy for realised volatility and search interest obtained from Google Trends is employed as the sentiment variable.

Within the econometric modeling framework, GARCH(1,1) and IGARCH(1,1) models are estimated as baseline models under two alternative innovation distributions: the Gaussian and the Student-t. The Student-t specification allows the conditional variance model to accommodate the heavy-tailed behaviour observed in cryptocurrency returns which is not captured by by the Normal GARCH models, and the sentiment effects are incorporated through the GARCH-X and IGARCH-X specifications, where lagged search interest enters

Setting	Value
Return variable	log_return
Realised volatility proxy	Squared returns (r_t^2)
Lagged sentiment variable	Search_Interest_lag
Benchmark model	GARCH_Normal
GARCH order (p, q)	$(1, 1)$
GARCH innovations compared	Normal, Student-t
RF baseline features	lag_sq_return_1, lag_sq_return_5, lag_abs_return_1, lag_return_1, lag_log_volume_1
RF sentiment-augmented features	lag_sq_return_1, lag_sq_return_5, lag_abs_return_1, lag_return_1, lag_log_volume_1, lag_sentiment_1, lag_sentiment_5
In-sample size	2,513
Out-of-sample size	629

Table 4.9: *Model specifications and estimation setup. The primary realised volatility proxy used for evaluation is squared returns, consistent with the GARCH-family forecasts generated in R.*

the conditional variance equation as an exogenous regressor.

For the machine learning framework, Random Forest regression models are used to forecast volatility with the baseline specification incorporating lagged volatility and return features, while the sentiment-augmented model adds lagged sentiment variables.

The dataset contains 3,142 observations in total from January 1, 2017 to September 8, 2025 and of these, 2,513 observations are used for in-sample estimation, while the remaining 629 observations are reserved for out-of-sample forecast evaluation.

4.4.2 Baseline econometric models

GARCH(1,1)-family models

Under Gaussian innovations, the baseline GARCH(1,1) in table 4.10, the ARCH parameter ($\alpha_1 = 0.1192$) and the GARCH parameter ($\beta_1 = 0.8402$) imply a persistence level of $\alpha_1 + \beta_1 = 0.9594$, indicating highly persistent volatility dynamics with both coefficients being strongly statistically significant with p-values of 0.0000. The conditional mean parameter ($\mu = 0.1858$) as well as the constant term($\omega = 0.6612$) are also statistically significant with

	GARCH Normal				GARCH Student-t			
Parameter	Coef.	Std.Err.	t-stat	p-value	Coef.	Std.Err.	t-stat	p-value
α_1	0.1192	0.0142	8.3671	0.0000	0.0857	0.0139	6.1765	0.0000
β_1	0.8402	0.0167	50.3722	0.0000	0.9133	0.0163	56.0578	0.0000
μ	0.1858	0.0552	3.3692	0.0008	0.1215	0.0411	2.9597	0.0031
ν	—	—	—	—	3.2411	0.1833	17.6805	0.0000
ω	0.6612	0.0996	6.6416	0.0000	0.1727	0.0801	2.1559	0.0311
AIC	5.2894				5.0655			
BIC	5.2971				5.0751			
Log-Likelihood	-8305.5939				-7952.8942			

Table 4.10: *Parameter estimates for baseline GARCH(1,1) models under Gaussian and Student-t innovations.*

p-values of 0.0008 and 0.0000 respectively. The significance of the mean parameter indicates that average returns are non-zero over the sample period, although this effect is typically small in magnitude relative to volatility dynamics, and the 0.0000 p-value for the constant term which reflects a positive baseline level of conditional variance, ensures that volatility remains strictly positive even in the absence of recent shocks respectively.

When Student-t innovations are assumed as in table 4.10, the ARCH coefficient declines to 0.0857, while the GARCH coefficient increases to 0.9133, yielding a persistence level close to unity ($\alpha_1 + \beta_1 = 0.9990$) with the estimated degrees-of-freedom parameter ($\nu = 3.2411$) significantly confirming heavy-tailed innovations. The intercept term ω decreases substantially under the heavy-tailed specification (0.1727 compared to 0.6612 under Gaussian errors), which means that part of the unconditional variance is absorbed by the fat-tailed distribution.

The model fit improves under Student-t innovations as shown by the increase in the log-likelihood from -8305.5939 under Gaussian errors to -7952.8942 under the Student-t specification, while both AIC and BIC values decline, indicating a better in-sample fit when heavy-tailed innovations are allowed.

	IGARCH Normal				IGARCH Student-t			
Parameter	Coef.	Std.Err.	t-stat	p-value	Coef.	Std.Err.	t-stat	p-value
α_1	0.1639	0.0168	9.7415	0.0000	0.0859	0.0132	6.4886	0.0000
β_1	0.8361	—	—	—	0.9141	—	—	—
μ	0.1802	0.0546	3.2998	0.0010	0.1211	0.0408	2.9661	0.0030
ν	—	—	—	—	3.2270	0.1337	24.1300	0.0000
ω	0.4935	0.0750	6.5836	0.0000	0.1674	0.0601	2.7836	0.0054
AIC	5.2975				5.0647			
BIC	5.3033				5.0724			
Log-Likelihood	-8319.3774				-7952.6269			

Table 4.11: *Parameter estimates for baseline IGARCH(1,1) models under Gaussian and Student-t innovations.*

IGARCH(1,1)-family models

In the IGARCH framework, persistence is constrained to unity by construction and under Gaussian innovations in table 4.11, the ARCH parameter is estimated at $\alpha_1 = 0.1639$ with the corresponding GARCH parameter $\beta_1 = 0.8361$ and the conditional mean parameter ($\mu = 0.1802$) and the variance intercept ($\omega = 0.4935$) are both statistically significant. The unit persistence implied by the IGARCH specification indicates that volatility shocks do not decay over time, but instead have a permanent effect on the conditional variance and as a result, the model does not exhibit mean reversion in volatility, meaning that periods of high or low volatility can persist indefinitely following a shock.

Under Student-t innovations, the ARCH parameter declines to 0.0859 while the GARCH parameter increases to 0.9141, which is a reflection of a similar trade-off between immediate shock response and long-run persistence as observed in the standard GARCH model. The estimated degrees-of-freedom parameter ($\nu = 3.2270$) again significantly indicates strongly heavy-tailed return innovations.

Consistent with the GARCH results, model fit improves under the Student-t specification. The log-likelihood increases from -8305.5939 under Gaussian errors to -7952.942 when Student-t innovations are used, while the information criteria decrease accordingly which confirms that allowing for heavy-tailed innovations better captures the distributional

characteristics of Bitcoin returns.

4.4.3 Sentiment-augmented econometric models

GARCH-X models

	GARCH-X Normal				GARCH-X Student-t			
Parameter	Coef.	Std.Err.	t-stat	p-value	Coef.	Std.Err.	t-stat	p-value
μ	0.1859	0.0553	3.3624	0.0008	0.1136	0.0403	2.8189	0.0048
γ (Sentiment)	0.0000	0.0033	0.0000	1.0000	0.0000	0.0020	0.0000	1.0000
ω	0.6621	0.1052	6.2925	0.0000	0.1156	0.0890	1.2994	0.1938
α_1	0.1193	0.0165	7.2358	0.0000	0.1064	0.0225	4.7302	0.0000
β_1	0.8400	0.0212	39.6748	0.0000	0.9189	0.0143	64.2452	0.0000
ν	—	—	—	—	2.8792	0.1838	15.6658	0.0000
Log-Likelihood	-8302.8688				-7947.4408			
AIC	5.2900				5.0643			
BIC	5.2996				5.0758			

Table 4.12: *Parameter estimates for sentiment-augmented GARCH-X(1,1) models under Gaussian and Student-t innovations.*

Across both innovation assumptions of the sentiment-augmented GARCH models in table 4.12, estimated sentiment coefficient (γ) is effectively zero with p-values of 1.0000, indicating that lagged search interest does not exert a statistically significant effect on conditional variance within the GARCH-X framework.

Importantly, the remaining variance parameters remain very similar to those obtained in the baseline GARCH models given that under the Gaussian innovations, the ARCH and GARCH parameters are $\alpha_1 = 0.1193$ and $\beta_1 = 0.8400$ respectively, implying strong volatility persistence. Under Student-t innovations, the persistence level remains close to unity with $\alpha_1 = 0.1064$ and $\beta_1 = 0.9189$ and the estimated degrees-of-freedom parameter ($\nu = 2.8792$) continues to confirm the heavy-tailed nature of Bitcoin returns.

The model fit is also largely unchanged relative to the baseline models as the log-likelihood values remain similar, suggesting that including sentiment does not materially improve in-sample fit within the linear GARCH framework.

IGARCH-X models

	IGARCH-X Normal				IGARCH-X Student-t			
Parameter	Coef.	Std.Err.	t-stat	p-value	Coef.	Std.Err.	t-stat	p-value
μ	0.1804	0.0546	3.3029	0.0010	0.1205	0.0411	2.9276	0.0034
γ (Sentiment)	0.0000	0.0032	0.0000	1.0000	0.0000	0.0019	0.0000	1.0000
ω	0.4940	0.0874	5.6534	0.0000	0.1678	0.0788	2.1293	0.0332
α_1	0.1641	0.0216	7.6109	0.0000	0.0860	0.0139	6.1946	0.0000
β_1	0.8359	—	—	—	0.9140	—	—	—
ν	—	—	—	—	3.2269	0.1355	23.8135	0.0000
Log-Likelihood	-8316.6915				-7950.2027			
AIC	5.2981				5.0654			
BIC	5.3058				5.0750			

Table 4.13: *Parameter estimates for sentiment-augmented IGARCH-X(1,1) models under Gaussian and Student-t innovations.*

Consistent with the GARCH-X results in table 4.13, the sentiment coefficient remains effectively zero across both innovation assumptions, with p-values equal to 1.0000 which implies that as in the case of the GARCH-X variations, lagged search interest does not contribute to explaining conditional variance within the IGARCH framework. Again, the remaining variance parameters are nearly identical to those estimated in the baseline IGARCH models with the ARCH parameter is $\alpha_1 = 0.1641$ with the corresponding GARCH parameter $\beta_1 = 0.8359$ under Gaussian innovations. The ARCH coefficient declines to 0.0860 while the GARCH coefficient increases to 0.9140 under the Student-t innovations, which also indicates highly persistent volatility dynamics.

These results indicate that the specific sentiment proxy used in this study, lagged Google Trends search interest, does not provide incremental explanatory power for conditional variance once past volatility dynamics are accounted for. This, however, does not imply that sentiment is irrelevant for cryptocurrency markets but rather, it suggests that the predictive content of behavioural variables depends critically on how sentiment is measured. In particular, Google Trends captures aggregate search intensity, which is more closely related to investor attention than to directional sentiment and as such, it may combine heterogeneous signals, including curiosity, speculation, and information demand, which do not necessarily

translate into a consistent effect on volatility.

4.4.4 Econometric models diagnostics

Residual diagnostics

Model	Lag	LB Stat (Resid)	p-value (Resid)	LB Stat (Sq. Resid)	p-value (Sq. Resid)
GARCH-Normal	10	16.3477	0.0120	6.1429	0.4074
	20	22.7527	0.1205	8.8035	0.9213
GARCH-Student-t	10	17.4817	0.0037	5.4926	0.3588
	20	24.6652	0.0546	10.2562	0.8033
IGARCH-Normal	10	17.4011	0.0079	6.9736	0.3233
	20	24.3291	0.0826	11.1744	0.7986
IGARCH-Student-t	10	17.5215	0.0036	5.4856	0.3595
	20	24.7241	0.0538	10.2721	0.8023

Table 4.14: *Ljung–Box diagnostics for baseline GARCH-family models. Tests are applied to standardized residuals and squared standardized residuals.*

Across all models, the Ljung–Box tests in table 4.14 applied to the squared standardized residuals are insignificant at both lag 10 and lag 20, with p-values ranging from approximately 0.32 to 0.92 which indicates that the conditional variance equations adequately capture the ARCH effects present in the data, leaving no significant autocorrelation in the second moment for all the models.

For the standardized residuals, however, some evidence of short-run autocorrelation is present at lag 10, with p-values well below 0.05 for all of the specifications. However, these effects largely disappear at lag 20, where the p-values increase above conventional significance thresholds of 5% which suggests that any remaining dependence in the conditional mean is weak and limited to very short horizons, thus, showing that the baseline GARCH-family models successfully absorb the volatility clustering observed in the raw returns, with no remaining serial dependence in squared residuals.

The results of the Ljung-Box diagnostics for the sentiment-augmented GARCH-X and IGARCH-X in table 4.15 mirror those obtained for the baseline models since in all specifications, the test results on the squared standardized residuals remain statistically

Model	Lag	LB Stat (Resid.)	p-value (Resid.)	LB Stat (Sq. Resid.)	p-value (Sq. Resid.)
GARCH-X Normal	10	16.4055	0.0058	6.1202	0.2947
	20	22.7661	0.0892	8.7819	0.8886
GARCH-X Student-t	10	18.4923	0.0010	5.2821	0.2596
	20	25.9263	0.0265	10.4887	0.7257
IGARCH-X Normal	10	17.4472	0.0037	6.9549	0.2240
	20	24.3341	0.0596	11.1579	0.7413
IGARCH-X Student-t	10	17.5972	0.0015	5.4622	0.2431
	20	24.7619	0.0370	10.2540	0.7434

Table 4.15: *Ljung–Box diagnostics for sentiment-augmented GARCH-family models.*

insignificant at both lag 10 and lag 20, with p-values generally above 0.22 and reaching as high as 0.89 which means that the inclusion of sentiment does not introduce variance misspecification.

As with the baseline models, some evidence of short-run autocorrelation appears in the standardized residuals at lag 10 for all the models and stronger, particularly for the Student-t specifications. However, these effects weaken substantially at lag 20, where the p-values exceed the 5% confidence threshold for all the Gaussian innovations and approach this threshold for the Student-t at 0.0265 for GARCH-X and 0.0370 for IGARCH-X.

ARCH-LM tests

Model	Lag Order	ARCH–LM Statistic	p-value
GARCH-Normal	10	6.1854	0.7995
GARCH-Student-t	10	5.4781	0.8570
IGARCH-Normal	10	7.0830	0.7176
IGARCH-Student-t	10	5.4686	0.8578
GARCH-X Normal	10	6.1790	0.8000
GARCH-X Student-t	10	5.2394	0.8746
IGARCH-X Normal	10	7.0710	0.7187
IGARCH-X Student-t	10	5.4622	0.8582

Table 4.16: *ARCH–LM tests for remaining conditional heteroskedasticity in standardized residuals of GARCH-family models.*

Across all specifications in table 4.16, the null hypothesis of no remaining ARCH effects

cannot be rejected since the reported p-values range from approximately 0.72 to 0.87, which implies that no significant conditional heteroskedasticity remains in the residuals after model estimation, hence confirming that the conditional variance equations successfully capture the volatility clustering present in the raw return series. Importantly, the diagnostic outcome is consistent across both the baseline and sentiment-augmented models, as well as across Gaussian and Student-t innovation assumptions at lag 10 which further confirms the adequacy of these econometric specifications.

4.4.5 Random forest model estimation

To provide a flexible nonlinear benchmark for volatility forecasting, Random Forest regression models are estimated using lagged volatility, return, and market activity variables as predictors with the dependent variable as the realised volatility proxied by squared returns, ensuring consistency with the volatility measure used in the econometric models and similar to the econometric models, two specifications, that is, baseline and sentiment-augmented Random Forests are considered. The baseline model includes lagged volatility dynamics, lagged returns, and trading activity variables while the sentiment-augmented specification extends this feature set by incorporating lagged Google trends search interest.

The model estimation is conducted using time-series cross-validation to preserve the temporal ordering of observations, implementing a five-fold rolling cross-validation scheme using the `TimeSeriesSplit` procedure and the hyperparameters are selected through grid search over the number of trees, maximum tree depth, and minimum leaf size. The candidate grid includes 300 and 500 trees, maximum depths of 4, 6, and 8, and minimum leaf sizes of 3 and 5 observations and the model selection is based on minimising mean squared prediction error within the cross-validation folds.

To ensure comparability with the econometric models, the Random Forest estimation sample is aligned with the same in-sample and out-of-sample split used in the GARCH-like models. Thus, the training sample contains 2,513 observations, while 629 observations are reserved for out-of-sample forecast evaluation.

In-sample fit

Model	MSE	RMSE	MAE	R^2
RF Baseline	2576.3414	50.7577	18.2999	0.1546
RF + Sentiment	2573.5543	50.7302	18.3459	0.1555

Table 4.17: *In-sample fit statistics for the Random Forest volatility models.*

The baseline RF specification as in table 4.17 produces a mean squared error (MSE) of 2576.34 and a root mean squared error (RMSE) of 50.76, with a mean absolute error (MAE) of 18.30 and an R^2 of 0.1546 which indicates that the model captures only a modest share of the variation in realised volatility, suggesting that a substantial portion of volatility dynamics remains unexplained within this specification.

Including sentiment variables leads to only marginal changes in the in-sample fit with the sentiment-augmented model just slightly reducing the MSE to 2573.55 and the RMSE to 50.73, while the MAE increases marginally to 18.35 and similarly, the coefficient of determination increases only slightly by 0.0009 to $R^2 = 0.1555$. Therefore, the overall differences between the two specifications are economically small, implying that sentiment variables contribute only limited additional explanatory power in the in-sample Random Forest estimation.

The weak in-sample fit of the Random Forest models suggests that the nonlinear tree-based specification is not extracting a strong additional signal from the selected predictors beyond what is already contained in recent volatility dynamics and the negligible difference between the baseline and sentiment-augmented specifications further indicates that lagged Google Trends search interest contributes very little incremental information within this framework.

Feature importance

In the sentiment-augmented specification, lagged volatility measures dominate the importance scores as shown in table 4.18, and the most influential predictors are the lagged absolute return (importance = 0.2101), the first lag of squared returns (0.2088), and

	RF Baseline		RF + Sentiment	
Feature	Importance	Rank	Importance	Rank
lag_sq_return_1	0.2214	3	0.2088	2
lag_sq_return_5	0.2511	1	0.1799	3
lag_abs_return_1	0.2345	2	0.2101	1
lag_return_1	0.1360	5	0.1214	4
lag_log_volume_1	0.1570	4	0.1155	5
lag_sentiment_1	–	–	0.0494	7
lag_sentiment_5	–	–	0.1149	6

Table 4.18: *Feature importance rankings for the Random Forest volatility models.*

the five-day lag of squared returns (0.1799). On the other hand, the sentiment variables contribute comparatively less to the predictive structure of the model with the five-day sentiment lag ranking sixth with an importance score of 0.1149, while the one-day sentiment lag contributes only 0.0494 and ranks seventh, with values that are noticeably lower than the importance scores associated with volatility and return dynamics.

In the baseline models, the lagged volatility variables again dominate the ranking, with the five-day squared return lag (0.2511) ranking first, followed by the lagged absolute return (0.2345), and the one-day squared return lag (0.2214) in the third place providing the largest contributions to predictive performance.

4.5 Out-of-sample forecasting performance

4.5.1 Forecast loss comparison

The out-of-sample forecasting performance of all the models is primarily ranked using the QLIKE loss functions as reported in table 4.19 in which lower values indicate better predictive accuracy. The baseline GARCH-Normal model achieves the lowest QLIKE loss (1.7459) and therefore serves as the benchmark specification, followed closely by the sentiment-augmented GARCH-X model with a QLIKE value of 1.7563, while the IGARCH-X Normal model ranks third with a loss of 1.7946 which shows that the differences among the top-performing econometric models are relatively small.

Model	QLIKE	Rank	Mean Loss Differential
GARCH-Normal	1.7459	1	0.0000
GARCH-X Normal	1.7563	2	-0.0104
IGARCH-X Normal	1.7946	3	-0.0487
IGARCH-Normal	1.7971	4	-0.0512
GARCH-Student-t	1.8021	5	-0.0562
IGARCH-Student-t	1.8102	6	-0.0643
IGARCH-X Student-t	1.8221	7	-0.0762
GARCH-X Student-t	1.8842	8	-0.1383
RF Baseline	1.9845	9	-0.2386
RF + Sentiment	1.9877	10	-0.2418

Table 4.19: *Out-of-sample volatility forecast performance measured using QLIKE loss. Mean loss differentials are reported relative to the GARCH-Normal benchmark.*

In contrast, the Random Forest models produce substantially higher forecast losses with the baseline Random Forest recording a QLIKE value of 1.9845, while the sentiment-augmented specification performs slightly worse with a loss of 1.9877, values that place the Random Forest models at the bottom of the ranking as further illustrated by the mean loss differentials reported in the final column where relative to the GARCH-Normal benchmark, the Random Forest models exhibit loss increases of about 0.24 which is almost double the worst performing GARCH variant(GARCH-X Student-t) with a mean loss differential of -0.1383. This means that the nonlinear machine learning approach does not translate into improved forecast accuracy for daily Bitcoin volatility within the evaluation sample.

Across the out-of-sample comparison, the results favour the econometric models over the Random Forest alternatives as the strongest performance is achieved by the standard GARCH specification under Gaussian innovations, while the RF models rank consistently below the GARCH-family forecasts. This pattern indicates that, for one-step-ahead Bitcoin volatility forecasting with squared returns as the evaluation proxy, explicitly modeling conditional heteroskedasticity and volatility persistence is more effective than relying on a flexible nonparametric regression alone.

4.5.2 Forecast dynamics

GARCH-family volatility forecasts

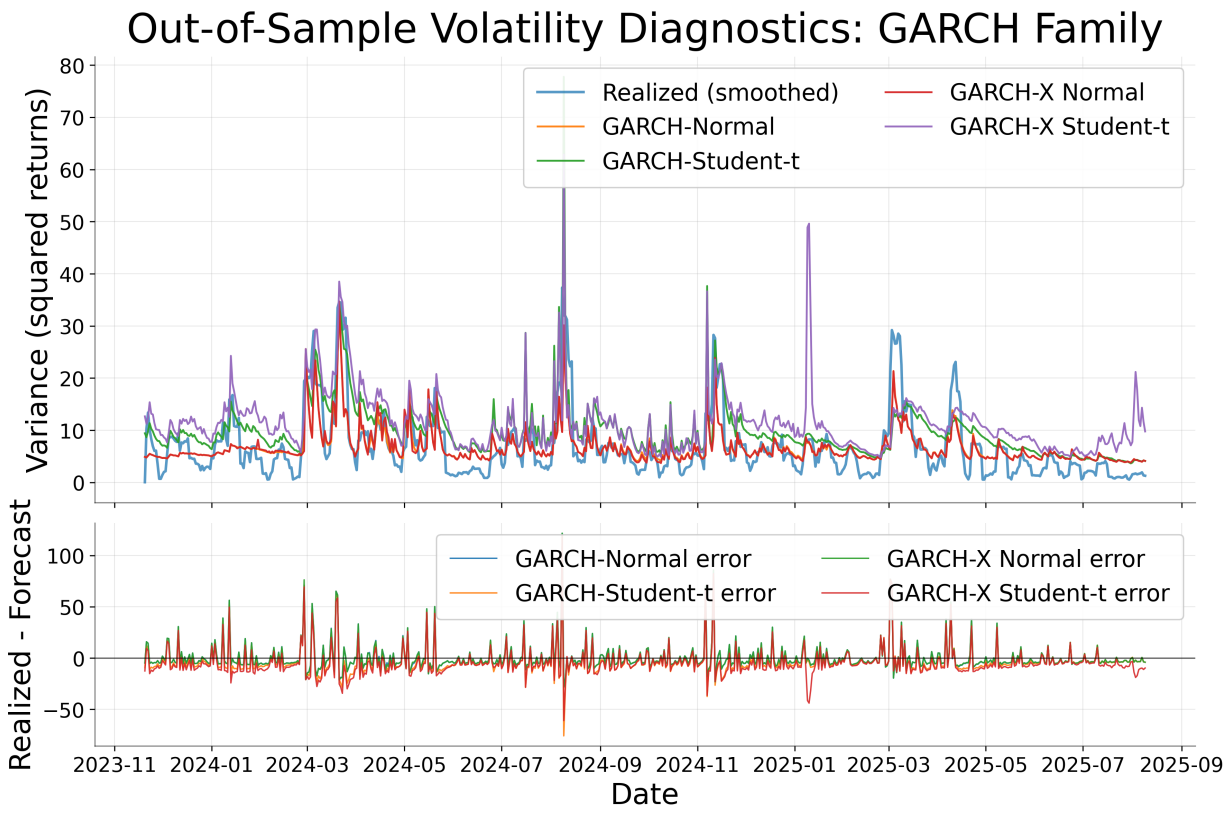


Figure 4.4: *GARCH and GARCH-X forecasts*

Figure 4.4 presents the out-of-sample volatility diagnostics for the GARCH-family models in which the upper panel displays the volatility forecasts generated by the different GARCH specifications together with the realised volatility proxy based on squared returns, while the lower panel shows the corresponding forecast errors defined as the difference between realised and predicted volatility.

The realised volatility series exhibits pronounced spikes and short-lived bursts of volatility, which affirms the clustering behaviour in cryptocurrency markets, and specifically, in Bitcoin. In this plot, the GARCH-family forecasts track the general level and persistence of volatility reasonably well with ocassional higher spikes as the models respond to periods of elevated volatility by gradually increasing the conditional variance forecasts and subsequently declining as market conditions stabilise, a feature of the autoregressive

structure embedded in the conditional variance equation.

The forecasts, however, appear noticeably smoother than the realised volatility series and tend to underreact to extreme spikes with the plot showing that several large volatility bursts, particularly around mid-2024 and early-2025, are only partially captured by the model forecasts.

The forecast errors in the lower panel tend to spike during periods of abrupt volatility increases, which indicates that the models underestimate the magnitude of extreme volatility shocks. Outside these periods, however, the errors remain relatively small and centred around zero, which shows that the models capture the general volatility level reasonably well, with relatively small differences between model specifications, though the Student-t based models occasionally generate slightly higher volatility forecasts during turbulent periods which reflects their heavier-tailed distributional assumption. Furthermore, the sentiment-augmented GARCH-X specifications display only minor deviations from the baseline GARCH forecasts, a confirmation for the earlier results in this study, indicating that the sentiment variable provides only limited additional explanatory power.

IGARCH-family volatility forecasts

Similar to figure 4.4, figure 4.5 shows the volatility forecasts generated by the IGARCH-family model specifications in the upper panel and the corresponding forecast errors in the lower panel. The volatility series, similarly, exhibit pronounced spikes and clear clustering behaviour, reflecting the episodic nature of volatility in cryptocurrency markets.

Due to the unit persistence constraint imposed in the IGARCH specification, however, volatility shocks tend to decay more slowly compared with the standard GARCH models and consequently, following major volatility spikes, the IGARCH forecasts remain elevated for longer periods even after the realised volatility begins to decline, producing smoother forecast dynamics and slower adjustments to changing market conditions. The forecast errors in the lower panel exhibit behaviour similar to the GARCH-family models in figure 4.4.

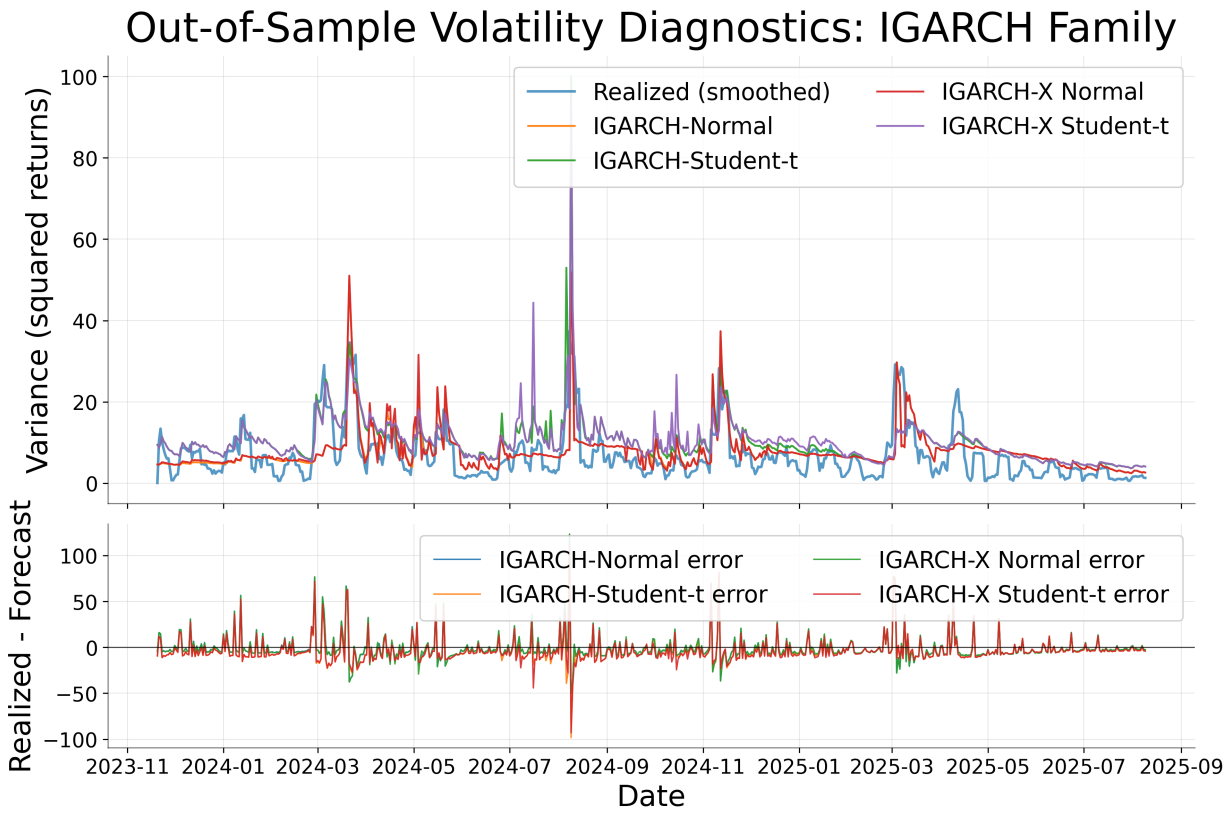


Figure 4.5: *IGARCH and IGARCH-X forecasts*

Random Forest volatility forecasts

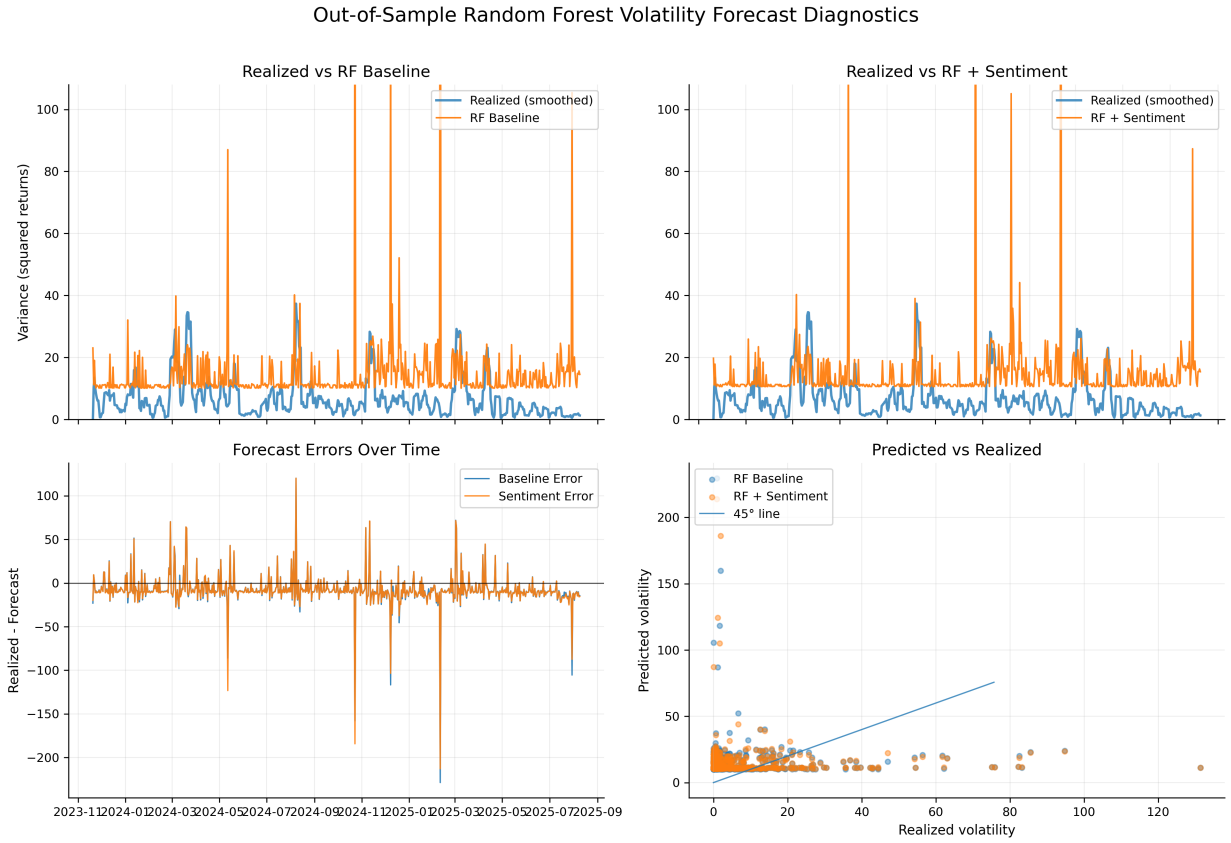


Figure 4.6: *RF models forecast diagnostics*

Figure 4.6 presents diagnostic plots for the out-of-sample volatility forecasts generated by the Random Forest models in which panels (a) and (b) compare the realised volatility proxy with the forecasts produced by the baseline Random Forest model and the sentiment-augmented Random Forest model, respectively.

Both models, despite tracking correctly the general direction of volatility movements exhibit substantially higher variability than the realised volatility series with both models generating sharper spikes compared to the GARCH-like models that exceed the magnitude of the observed volatility, particularly during periods of market turbulence.

In panel (c), the forecast errors overtime are shown, and even though most errors fluctuate around zero, the distribution is characterised by occasional large positive and negative deviations, which aligns with the substantial overprediction or underprediction of volatility particularly during extreme market episodes.

Panel (d) plots predicted volatility against realised volatility where dispersion of points away from the 45-degree line indicates considerable forecast noise, with many observations clustered at relatively low realised volatility levels but associated with higher predicted volatility which suggests that the Random Forest models tend to overreact to certain predictor signals and generate exaggerated forecasts even during otherwise moderate volatility conditions. Here, the baseline and the sentiment-augmented models produce very similar forecast paths throughout the evaluation period and this visual similarity confirms the results of the feature importance analysis which showed that sentiment variables contribute relatively little predictive information relative to lagged volatility measures.

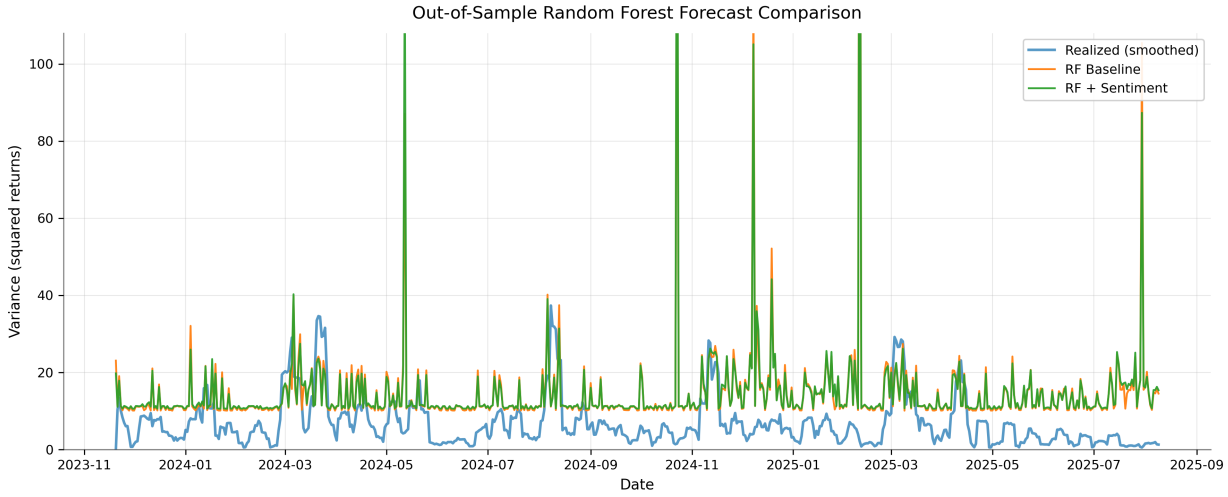


Figure 4.7: *Random baseline vs sentiment models*

In figure 4.7, a visual comparison of the Random Forest models reveals that both models generate volatility forecasts that move closely together over the entire out-of-sample period, with the forecasts exhibiting near-identical responses to the volatility spikes. In particular, most large forecast spikes occur simultaneously in both models, suggesting that they are primarily driven by the underlying return dynamics rather than by additional information contained in the sentiment measure.

4.6 Robustness checks

4.6.1 Superior predictive analysis(SPA) test

Statistic	Value
SPA Statistic	-0.2608
p-value	0.5700
Observations	629
Models Tested	9
Benchmark Model	GARCH-Normal
Block Length	10
Bootstrap Replications	1000

Table 4.20: *Superior Predictive Ability (SPA) test results relative to the GARCH-Normal benchmark model.*

The results of the Superior Predictive Ability (SPA) test are reported in table 4.20 evaluates whether any of the competing forecasting models significantly outperforms the chosen benchmark model, which in this case is the GARCH-Normal specification. The reported SPA statistic is negative (-0.2608) with an associated bootstrap p-value of 0.57 based on 1000 bootstrap replications and a block length of 10. Since the p-value is well above conventional significance levels, the study fails to reject the null hypothesis that the benchmark model is not inferior to the competing models, meaning that none of the alternative forecasting models provides a statistically superior predictive performance in comparison to the GARCH-Normal benchmark when evaluated using the chosen loss function. Although some models may exhibit marginal improvements in point forecast accuracy, these differences are not large enough to be considered statistically significant within the SPA testing framework.

Illustrated in figure 4.8 is the bootstrap distribution of the SPA statistic obtained from the resampling procedure in which the histogram shows the empirical distribution generated from the bootstrap replications, while the vertical dashed line represents the observed SPA statistic computed from the original sample. The observed statistic lies well within the central region of the bootstrap distribution rather than in the extreme tail which provides a visual evidence that is consistent with the relatively large p-value reported in table 4.20 and

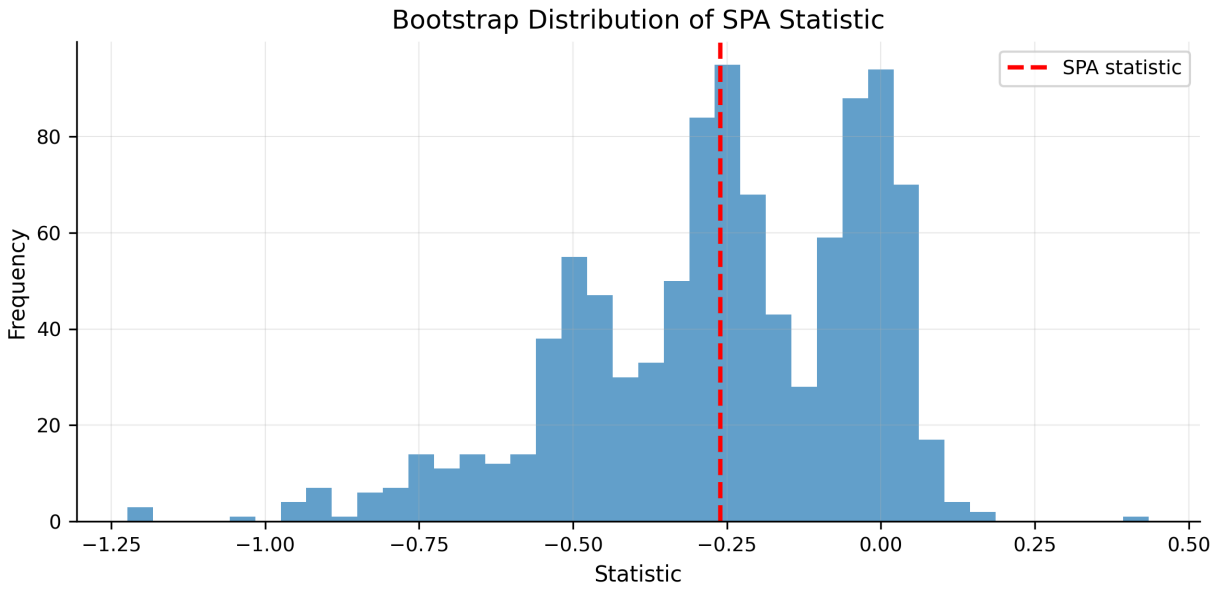


Figure 4.8: *Caption*

further supports the conclusion that there is no statistically significant evidence that any of the competing models outperform the benchmark model.

4.6.2 Model confidence set(MSC)

Model	Included in Final MCS
GARCH-Normal	Yes
GARCH-Student-t	Yes
GARCH-X Normal	Yes
GARCH-X Student-t	Yes
IGARCH-Normal	Yes
IGARCH-Student-t	Yes
IGARCH-X Normal	Yes
IGARCH-X Student-t	Yes
RF Baseline	Yes
RF + Sentiment	Yes

Table 4.21: *Model Confidence Set inclusion results based on the out-of-sample loss matrix.*

Table 4.21 reports the inclusion results from the Model Confidence Set (MCS) procedure, based on the out-of-sample loss matrix, which evaluates whether any forecasting model can

be statistically eliminated from the set of superior models according to the selected loss function.

The results indicate that all ten models remain included in the final Model Confidence Set, meaning that none of the competing models can be statistically rejected as inferior relative to the others at the chosen confidence level. In other words, although individual models may exhibit differences in forecast accuracy, these differences are not sufficiently large enough to justify excluding any model from the set of potentially best-performing specifications which suggests that both the traditional econometric models and the Random Forest models forecasting performance across model classes is broadly comparable within the evaluated sample period.

To further confirm this, the summary of the final MCS at the 90% confidence level in table 4.22 retains all the ten forecasting models which means that the predictive performance differences among the competing models are statistically indistinguishable within the sample, which also justifies the SPA results in table 4.20 which similarly failed to identify any model with significantly superior predictive ability relative to the benchmark.

Confidence Level	Models Retained	Final Model Confidence Set
90%	10	GARCH-Normal, GARCH-Student-t, GARCH-X Normal, GARCH-X Student-t, IGARCH-Normal, IGARCH-Student-t, IGARCH-X Normal, IGARCH-X Student-t, RF Baseline, RF + Sentiment

Table 4.22: *Summary of the final Model Confidence Set (MCS) at the 90% confidence level.*

4.6.3 Alternative volatility proxy

Absolute returns

The forecast evaluation results in table 4.23 are obtained using absolute returns as an alternative volatility proxy, and since the GARCH and IGARCH models generate forecasts for conditional variance, the forecasts are converted to volatility by taking square roots to ensure comparability with the absolute-return proxy.

Model	MSE	RMSE	MAE	Rank	Mean Loss Differential
GARCH-Student-t	3.7396	1.9338	1.6149	1	0.0000
GARCH-X Student-t	3.7637	1.9400	1.6137	2	-0.0242
IGARCH-Student-t	4.2515	2.0619	1.7119	3	-0.5119
IGARCH-X Student-t	4.2655	2.0653	1.7154	4	-0.5259
GARCH-Student-t	4.8441	2.2009	1.8798	5	-1.1045
IGARCH-Student-t	4.9916	2.2342	1.9139	6	-1.2520
IGARCH-X Student-t	5.1015	2.2586	1.9320	7	-1.3619
GARCH-X Student-t	5.6851	2.3843	2.0683	8	-1.9455
RF Baseline	7.4436	2.7283	2.3373	9	-3.7040
RF + Sentiment	7.4622	2.7317	2.3446	10	-3.7226

Table 4.23: *Forecast evaluation under the alternative volatility proxy based on absolute returns. Model forecasts originally produced on the variance scale are converted to volatility by taking square roots.*

The results show that the overall ranking of the models remains broadly consistent with the baseline evaluation as the GARCH-Normal model achieves the best forecasting performance across the loss measures, followed closely by the sentiment-augmented GARCH-X Normal specification and the IGARCH models perform moderately well but remain slightly less accurate than the standard GARCH variants. Consistent with the case of relative volatility, the Random Forest models exhibit weaker performance under this alternative volatility proxy, ranking ninth and tenth respectively which further suggests that the machine learning models may be more sensitive to the choice of volatility proxy compared with the parametric GARCH-family models.

Alternative evaluation window (Restricted OOS)

The forecast evaluation results under the restricted out-of-sample evaluation window as in table 4.24 where early portion of the forecasting period is excluded to examine the robustness of the model rankings shows that under RMSE, the results remain broadly similar to the primary evaluation criteria, QLIKE. The GARCH-Normal model continues to achieve the best overall performance while the Random Forest models consistently maintian the worst performance with substantially large forcast errors compared to the parametric volatility models. Similarly, the inclusion of of the sentiment variable does not materially improve the performance of all the models except for GARCH-X Normal which ranks first under MAE,

Model	RMSE	MAE	RMSE Rank	MAE Rank	RMSE Loss Differential
GARCH-Student-t	12.7346	7.0477	1	2	0.0000
GARCH-X Student-t	12.7466	7.0132	2	1	-0.0120
IGARCH-X Student-t	13.4837	7.8676	3	3	-0.7492
IGARCH-Student-t	13.4914	7.8867	4	4	-0.7568
GARCH-Student-t	13.8251	8.6135	5	5	-1.0905
IGARCH-Student-t	14.2929	8.9795	6	6	-1.5583
IGARCH-X Student-t	14.3230	9.1142	7	7	-1.5884
GARCH-X Student-t	14.4595	9.6804	8	8	-1.7249
RF + Sentiment	21.3354	13.2051	9	9	-8.6008
RF Baseline	21.6197	13.2073	10	10	-8.8851

Table 4.24: *Forecast performance under a restricted out-of-sample evaluation window.*

outperforming the baseline GARCH-Normal by barely 0.0345.

Robustness of sentiment effects

Model	RMSE	MAE	RMSE Rank	MAE Rank	RMSE Loss Differential
GARCH-Student-t	13.4567	7.8006	1	1	0.0000
GARCH-X Student-t	13.5244	7.8011	2	2	-0.0678
IGARCH-Student-t	14.1781	8.4996	3	3	-0.7214
IGARCH-X Student-t	14.2116	8.5309	4	4	-0.7549
GARCH-Student-t	14.3912	9.4739	5	5	-0.9345
IGARCH-Student-t	14.7127	9.7388	6	6	-1.2560
IGARCH-X Student-t	14.7434	9.8381	7	7	-1.2868
GARCH-X Student-t	15.1431	10.7216	8	8	-1.6865
RF Baseline	20.3581	12.9914	9	9	-6.9014
RF + Sentiment	20.4122	13.0253	10	10	-6.9555

Table 4.25: *Forecast performance when the evaluation sample is restricted to observations where the sentiment variable lagged by an additional day is available.*

To examine whether the predictive contribution might be sensitive to the timing of the the information, I evaluate and report the forecast evaluation results in table 4.25 when the sentiment variable is lagged by an additional day, and the results remain consistent with the baseline forecast evaluation as the GARCH-Normal model continues to achieve the lowest RMSE and MAE values, followed closely by the sentiment-augmented GARCH-X Normal specification. Given that this trend is consistent for the IGARCH variants as well as the Random Forest models, the sentiment variations of the models are proven to have insignificant predictive power contribution even when alternative lag structures of the sentiment variable are considered.

Chapter 5: Discussion

5.1 Interpretation of findings

5.1.1 Stylised facts of Bitcoin volatility

The empirical results confirm that Bitcoin volatility exhibits the key stylised facts commonly observed in financial time series, including volatility clustering, strong persistence, and pronounced leptokurtosis. In particular, the estimated persistence parameters from the GARCH(1,1) models, with $\alpha_1 \approx 0.12$ and $\beta_1 \approx 0.84$, imply a persistence level close to unity, indicating that volatility shocks decay slowly over time. This high degree of persistence is consistent with the autoregressive structure of conditional variance documented in the ARCH and GARCH literature.

These features are central to understanding the model comparison results reported in this study and the strong volatility clustering and persistence observed in Bitcoin returns favour econometric models such as GARCH, which are explicitly designed to capture time-varying variance dynamics. By contrast, models that do not impose a volatility structure, such as Random Forests, may struggle, as is the case in this study, to fully account for these persistent conditional variance patterns.

The distributional properties of the residual point to the presence of fat tails that implies that the extreme movements in Bitcoin returns occur more frequently than would be predicted under normal distribution which is in line with the finding of Bollerslev et al. (1994) for the wide financial econometric filed, and has been widely documented in key cryptocurrency studies, for example by Chu et al. (2017), Katsiampa (2017) and Ardia et al. (2019). On the other hand, the superior performance of the Gaussian model specification relative to the Student-t specifications in this study questions the importance of accommodating heavy-tailed behaviour when modeling Bitcoin volatility.

The evidence on the asymmetry appears to be weaker or less consistent than in traditional

equity markets and while volatility responses are well documented for equeties (Nelson, 1990), the results in this study shows that such effects are either absent or less pronounced in Bitcoin markets and in line with the findings of the previous studies as Katsiampa (2017) and Cheikh et al. (2020) who, both reported mixed or limited evidence of leverage effects in cryptocurrency returns.

The results also show that Bitcoin volatility may exhibit periods of structural instability and regime-dependent behaviour, and compared to the traditional asset classes, Bitcoin is characterised by rapid changes in market participation, evolving regulatory frameworks and shifts in investor composition, features which can lead to time variation in volatility dynamics as documented by Dyhrberg (2016) and Ardia et al. (2019). Thus, the relatively high persistence observed in the estimated models may, therefore, prartly capture underlying regime shifts rather than the purely stationary volatility dynamics.

5.1.2 Interpretation of econometric model results

One of the most striking features of the econometric models results is the consistently high degrees of persistence in the conditional variance across all the GARCH specifications where the sum of the ARCH and GARCH parameters is consistently close to unity, suggesting that the shocks to volatility tend to dissipate only gradually, a finding more in line with the results from Engle & Bollerslev (1986) for the financial time-series data. At the same time, the persistence observed here may be somewhat overstated in the sense that it could be absorbing the structural changes in the underlying data-generating process rather than the purely stable dynamics.

The IGARCH models specifications, which by construction, impose a unit root in the model volatility process further re-enfornces the point of high persistence as these models effectively assume that the shocks have a permanent effect. While the IGARCH models can sometimes provide a better fit in the highly persistent environments, the results in this study show that IGARCH does not offer a clear advantage over the standard GARCH models, a finding in contradiction to Chu et al. (2017). In fact, the forecasting performance of the IGARCH models are weaker in several cases, as the best IGARCH model, IGARCH-X Normal ranks third in the table 4.19, which may indicate that imposing full persistence is

too restrictive for this dataset. In other words, this study suggests that while volatility in Bitcoin is highly persistent, it is perhaps not quite to the extent implied by the integrated processes.

The comparison between the Gaussian and the Student-t innovations point to no improvement whatsoever given that all the Student-tt innovations rank below all their corresponding Gaussian variations as in table 4.19 when heavy tails are explicitly accounted for, which raises the possibility that while fat tails are clearly present in the data, their specific practical importance may depend more on the evaluation metric and sample period.

5.1.3 Interpretation of sentiment effects

The inclusion of Google Trends as an exogenous regressor in the GARCH-X and IGARCH-X specifications produces weak and statistically insignificant effects in most cases, and the same variable adds almost no explanatory power in the Random Forest framework, findings that suggest that the search-based attention proxy used in this study does not provide a strong or stable signal for one-step-ahead Bitcoin volatility forecasting once past volatility information is already included. This does not imply that behavioural variables are irrelevant in cryptocurrency markets, but rather, it suggests that the specific proxy employed here may be too coarse to isolate the type of sentiment that matters for volatility formation. In the Random Forest models, for example, the dominant predictors remain lagged volatility and return-based variables, while the inclusion of Google Trends leaves model fit almost unchanged.

A possible interpretation is the fact that Google Trends measures attention more than sentiment in the strict sense given that a rise in search intensity can reflect curiosity, confusion, media exposure, panic, or speculative excitement, all of which may have different implications for subsequent volatility. As a result, the same increase in search activity need not correspond to a consistent directional effect in the variance equation. This possibility is in line with the broader literature given that Mai et al. (2018) show that not all online signals are equally informative as the predictive content comes primarily from the "silent majority," while more visible online activity is not necessarily the most useful. Sapkota (2022) also finds that news-based sentiment can improve volatility forecasting, but this is

based on textual sentiment extracted from a large news corpus rather than on a broad attention proxy and more specifically, Hoang & Vo (2024) also show that Google search volume predicts returns, volatility, and trading volume, but their paper interprets Google searches as investor attention and price pressure rather than as a clean measure of sentiment itself.

The present results of this study, therefore, fit a more balanced reading of the literature, highlighting the fact that behavioural variables can matter, but their usefulness depends critically on how they are measured as search-based attention may be too aggregated to capture the qualitative differences across users, motives, and information environments that appear to drive stronger predictive effects in studies based on forums, news text, or richer nonlinear sentiment indices. This helps explain why Google Trends does not materially improve either the econometric models or the Random Forest specifications in this study.

5.1.4 Interpretation of forecasting performance

Bitcoin volatility displays pronounced persistence and clustering, the exact dynamics for which the GARCH models are designed to represent directly through the conditional variance equation. By contrast, Random Forests do not impose a volatility structure, but they must infer such persistence indirectly from lagged features, which can be less effective when the underlying process is dominated by stable heteroskedastic dynamics rather than by complex nonlinear interactions alone.

Importantly, this indicates that a relatively simple standalone Random Forest, as used in this study, is not sufficient to outperform well-specified econometric volatility models in this setting, a finding supported by prior literature as in Amoanu (2025)'s comparative study which outlines that ARMA-GARCH achieves the lowest forecast errors, while Random Forest and XGBoost are competitive but do not fully account for volatility dynamics in the same way as GARCH-type models. The findings of this study, thus, are more in line with the findings of Amoanu who explicitly notes that RF captures nonlinear mean relationships but lacks a direct mechanism for volatility clustering, which is closely aligned with the pattern observed in this study.

At the same time, the broader machine-learning literature is more mixed than a simple

"*Machine Learning models beat GARCH models*" narrative would suggest as Basher & Sadorsky (2022) report strong performance for Random Forests in forecasting Bitcoin price direction, but their target is directional movement rather than volatility. Similarly, Diane & Brijlal (2024) also find that RF outperforms ANN, but again the comparison is between two machine-learning models rather than between RF and a strong GARCH benchmark. In contrast, the papers that report the largest gains over traditional models typically rely on more sophisticated architectures, especially hybrid GARCH–LSTM designs rather than standalone tree-based models as is the case in Leber & Egyed (2025) who report superior performance for a sentiment-augmented GARCH-LSTM framework, and similar conclusions appear in other hybrid deep-learning studies.

The implication, therefore, is that the weak Random Forest performance in this thesis is not an anomaly, but is rather consistent with the view that volatility forecasting rewards models that encode persistence and conditional variance dynamics explicitly. Where machine learning succeeds most strongly in the literature, it often does so either on different targets or within hybrid systems that retain a GARCH backbone rather than replacing it altogether.

5.1.5 Interpretation of robustness analysis

When absolute returns are used in place of squared returns as an alternative proxy for volatility, the relative ranking of the models remains the same, with the GARCH-type models continuing to dominate under the standard error-based measures, consistent with the findings of Corsi (2009) who emphaised that models grounded in the conditional variance structure are well suited to capture the persistence in volatility. Consistently, the Random Forest models continue to perform relatively poorly which further suggests that their flexibility does not compensate for the weaker alignment with the underlying volatility process.

Allowing sentiments to enter the models with a delay leads only to marginal changes in model performance, but does not improve the relative rankings of the sentiment-augmented models, which is broadly in line with the results from Sapkota (2022) and later in Wang et al. (2023) who suggest that the sentiment effects may be present but are often unstable or dependent on specific modeling choice.

Under reduced out-of-sample window, the relative ordering of the models remains largely

the same which confirms that the models are stable in their respective forms. Therefore, as noted by P. R. Hansen (2005) and later extended by P. R. Hansen et al. (2011) that the robust model comparison should account for sampling variation and model uncertainty, the fact that the ranking does not change materially across the sub-samples suggests that the superior performance of the GARCH models is not driven by a particular segment of the data.

5.2 Comparison with existing literature

5.2.1 Relation to classical volatility modeling literature

The empirical findings of this study can be understood to be an extension of the classical volatility modeling framework developed in ARCH by Engle (1982) and GARCH by Bollerslev (1986) lietrature as most of the patterns observed in the Bitcoin volatility, most noteably persistence and clustering, are in alignment with the stylized facts that originally motivated these models. In this sense, therefore, the results show that even in new and rapidly evolving markets, the basic mechanisms underlying volatility appear to resemble those found in more established financial assets. That said, some aspects of the results are less in line with the traditional framework, for example, while high persistence is a heavily-present feature of the financial timeseries(Engle & Bollerslev, 1986), the near-integrated behaviour observed in some of the estimated models where the ARCH and GARCH parameters are very close to unity raises the question whether the standard GARCH model specifications, despite being the best performing models in this study, are fully adequate to capture the volatility of Bitcoin.

In the equity markets, negative shocks are often associated with larger increases in volatility than positive shocks(Nelson, 1990), but in this study, for the case of Bitcoin, the evidence for such leverage effects appear weaker and less consistent, which may suggest that some of the mechanisms underlying volatility in traditional financial markets may not directly translate equal performance and reliability for digital assets.

As highlighted by Engle & Bollerslev (1986), allowing for heavy-tailed innovations should

improve the statistical fit for volatility models by accommodating the excess kurtosis in returns. The results obtained in this study, albeit being broadly consistent with this insight, show that the gains in forecasting performance remains relatively modest, meaning that even though there is evident improvement in the in-sample fit, there is no significant gains in the out-of-sample forecasts. From the fact that the GARCH models are the strongest performers in this study, the study concludes that the classical framework is not merely described adequately adequate, but also well-equipped to forecast the volatility of Bitcoin, and by extension, digital assets.

5.2.2 Comparison with cryptocurrency volatility studies

A common resutls across many studies is that GARCH-type models provide reasonable description of Bitcoin volatility dynamics as documented by both Dyhrberg (2016) and Chu et al. (2017) who both found strong volatility clustering and persistence, and highlight that the standard GARCH models perform well relative to the other alternative models. Further works in this regard by Katsiampa (2017) and later Ardia et al. (2019) show that extensions of the GARCH framework are capable of capturing key features of cryptocurrency returns, and the results obtained in this study aligns well with this, given the dominance of the GARCH models over all the other specifications.

Unlike studies that identify gains from component, smooth-transition, regime-switching, or asymmetric GARCH variants as in Chu et al. (2017), the present thesis finds that a parsimonious Gaussian GARCH(1,1) provides the best out-of-sample volatility forecasts among the models considered. In particular, neither IGARCH specifications nor Student-t innovations improve forecasting accuracy despite their stronger persistence and superior in-sample fit. This difference highlights that the relative performance of volatility models depends critically on the forecasting objective, the evaluation metric, and the degree of model complexity employed.

5.2.3 Comparison with sentiment and investor attention literature

The results of this study provide only limited support for search-based attention as a useful volatility predictor in the specific form considered here. This differs from parts of the behavioural finance and cryptocurrency literature, although this difference is narrow given that Mai et al. (2018) show that the predictive content of online activity depends on user heterogeneity, and more specifically that signals from the silent majority are informative, whereas more visible activity from highly active users is less useful. Their findings imply that aggregate attention measures can conceal important differences in information quality across users and platforms, and in that sense, the weak results obtained with Google Trends in this thesis are not inconsistent with Mai and co-authors, but rather, they support the idea that a broad aggregate attention measure may be too blunt to capture the relevant behavioural mechanism.

A similar qualification applies to the wider sentiment literature as Sapkota (2022) finds that news-based sentiment improves HAR-RV volatility forecasts and that different sentiment measures operate at different horizons, but those results rely on textual sentiment extracted from 17,490 newspaper articles using sentiment dictionaries which is a much richer informational object than the unidimensional search-intensity series used here. In a later study, Hoang & Vo (2024) find that abnormal Google search volume predicts higher returns, volatility, and trading activity, yet their interpretation is explicitly attention-based and linked to price-pressure effects and their results, therefore, show that search intensity can matter in some settings, but they do not imply that Google Trends will always improve volatility forecasting once lagged variance dynamics are controlled for.

More recent studies also support the idea that predictive gains depend on the quality and structure of the sentiment measure as in Leber & Egyed (2025) who find that sentiment can improve nonlinear volatility forecasting, but they also report that Twitter-based sentiment indices contain substantially more predictive information than news-based indices in their setting. This, again, points to a key discovery for the present thesis that behavioural information is not homogeneous, and the forecasting value of sentiment variables depends heavily on the source, granularity, and the nonlinear transformation of the underlying data.

Against this background, therefore, the failure of Google Trends to improve the present models is not to be interpreted as a rejection of behavioural effects altogether, but as evidence that this particular proxy is too noisy or too aggregated for the forecasting task at hand.

5.2.4 Relation to machine learning forecasting studies

The machine-learning results in this thesis, albeit not being outside the range of evidence in the literature, are weaker than those reported in several recent studies, though they are not outside the range of evidence in the literature, thus a useful starting point is to distinguish between the different comparisons given that, some papers compare Random Forest with simpler statistical or machine-learning benchmarks and find that RF performs well. For instance, Basher & Sadorsky (2022) show that Random Forest and bagging outperform logit models in forecasting Bitcoin price direction, while Diane & Brijlal (2024) find that RF outperforms ANN in forecasting realised volatility in South Africa. These studies establish that RF can be a strong benchmark in some contexts, but neither implies that RF should systematically dominate well-specified GARCH-family volatility models.

On the other hand, a smaller but important set of studies shows that GARCH-type models can remain as good as or better than tree-based ML methods when volatility dynamics are central to the forecasting problem and in this regard, Amoanu (2025)'s comparative study is especially relevant given that the author finds that ARMA-GARCH achieves the lowest forecast errors overall, while Random Forest and XGBoost are competitive but do not model volatility persistence directly. The paper further notes that residual variance remains heteroskedastic under RF, reinforcing the view that nonlinear tree ensembles may still leave key volatility structure unmodeled. This closely matches the present findings, where GARCH and IGARCH dominate RF under QLIKE and remain superior on standard error metrics as well.

Machine learning models seems to perform best in their hybrid versions which incorporate the econometric models and the papers that report the strongest forecasting gains from machine learning generally do so through more sophisticated hybrid or deep-learning architectures rather than through standalone RF models as is the case in Leber & Egyed (2025) who report superior volatility and VaR performance for a sentiment-augmented

GARCH-LSTM system. The other related hybrid studies in the literature similarly combine econometric volatility structure with flexible nonlinear learners, and thus, the broader implication is that machine learning tends to perform best when it complements, rather than replaces, the econometric modeling of conditional variance. Viewed in this light, therefore, the present thesis contributes by showing that a simple RF implementation is not enough to outperform standard GARCH models for Bitcoin volatility over this sample period, and that stronger ML performance may require hybridisation, richer features, or both.

5.3 Limitations

5.3.1 Data limitations

Due to the absence of directly observed volatility, the study uses squared returns as the measure of realized volatility. Although this is the standard practice in empirical finance, particularly within the ARCH and GARCH literature, where volatility is treated as a latent process inferred from return innovations, e.g., (Engle, 1982; Bollerslev, 1986), as well as in more recent studies, e.g., (Sapkota, 2022; Zahid et al., 2022), that rely on such proxies when high-frequency realised measures are unavailable, volatility is inherently unobservable(Patton & Sheppard, 2009) and any proxy introduces measurement errors that can affect both the model estimation and evaluation, and the squared returns, in particular, can be quite noisy and may not fully capture the underlying volatility process especially at lower frequencies. A solutuon to this could possibly be the use of alternative measures such as realized volatility constructed from high frequency data as proposed by Corsi (2009) which provide more accurate representation, though such data are not always readily available over long sample periods.

The use of daily observations strikes a balance between availability and computational feasibility but it may obsecure the very important intra-day dynamics given that the volatility in cryptocurrency markets can evolve rapidly within a single day, and yet these short-term fluctuations are not captured in the daily open and close price points. As a result, aspects

of volatility behaviour related to market microstructure or high frequency trading is easily overlooked.

Bitcoin markets have undergone substantial changes over time, including shifts in market participation with the involvements of governments in buying and selling Bitcoin, trading infrustructure and regulatory attention which may mean that the sample period introduces certain constraints. As noted by Babalos et al. (2025), such structural developments can alter volatility dynamics while the regulatory perspectives, as in Gensler (2024), may influence market behaviour through changes in expectations and uncertainty. Similarly, Argente & Van Patten (2024) emphasize the increasing integration of the cryptocurrency markets with the broader financial market which is hinted to introduce new sources of volatility, thus, suggesting that the underlying data generating process is unlikely to be stable across the entire sample.

Even though Bitcoin price data are generally reliable, they may compromise the quality of the study given that they are sourced from decentralised market that operates across multiple exchanges with varying levels of liquidity and regulation. This can introduce discrepancies in the reported price and volumes traded, particularly during periods of market stress, and even though such issues are unlikely to overturn the findings of this study, they may have contributed to additional noise which can affect both the econometric estimations as well as the performance of the Random Forest models given the heavy dependence and sensitivity of the machine learning models on the data quality and feature constriction(Dudek et al., 2024). Similarly, the focus on single asset limits the generalisability of the findings given that Bitcoin is treated as a representative of broader cryptocurrency market even though it differs from the other digital assets in terms of liquidity, market maturity and investor composition.

5.3.2 Sentiment measurement limitations

In addition to general data considerations, there are also limitations related to how the sentiment variable is measured given that it is not always clear what type of sentiment is being captured, and as Mai et al. (2018) points out, search activity may reflect increased engagements with the market, but this engagement is not necessarily directional, as a

rise in search intensity could indicate optimism, concern, curiosity, or simply increased media attention. This lack of distinction between positive and negative sentiments makes it difficult to establish a stable relationship between the sentiment variable and the volatility outcomes. Related to this is the fact that search-based measures capture attention rather than behaviour, and while attention may precede trading activity, the connection between the two is not guaranteed owing to the fact that not all individuals who search for information participate in the market(Coulter, 2022) and those who do may respond in different ways.

5.3.3 Market structure considerations

Unlike the traditional financial markets which have developed over decades within the relatively stable institutional frameworks, cryptocurrency markets reamin in a process of rapid transformation which makes it more difficult to treat observed relationships as stable over time.

The composition of the Bitcoin submarket of the cryptocurrency markets has been evolving over the years, and in its early stages, Bitcoin trading was largely dominated by retail investors and speculative traders who contributed to the high volatility witnessed in the Bitcoin prices as in Dyhrberg (2016) who highlighted that the market was constantly reacting to new information. More recently, however, there has been increased institutional participation and the introduction of more sophisticated financial instruments, as reported by Ardia et al. (2019), changes that are likely to affect both the level and the dynamics of volatility as different types of participants may respond to information in different ways. The result of this is that the patterns that we observed in the earlier periods may not carry over directly to the more recent market conditions.

Regulatory developments have also provided another source of structural changes owing to the ongoing discussions around the classification, oversight and the integration of cryptocurrencies into the broader financial systems which have intensified over the years. This evolving regulatory framework can alter, as Gensler (2024) notes, the market behaviour by shaping expectations, reducing uncertainty in some cases, or introducing new forms of risk in others, developments which may contribute to the shifts in volatility that are not easily captured by models assuming stable environments.

The cryptocurrency markets have also not fully matured in comparison to the equity markets, for example, as shown by both Chu et al. (2017) and Katsiampa (2017), the early cryptocurrency markets exhibited strong volatility clustering and limited evidence of stabilising mechanisms, and despite the increased liquidity and improved infrastructure leading to more efficient price discovery over time, more recent eividence as in Cheikh et al. (2020) suggests that volatility remains elevated and sensitive to external shocks. This combination of maturation and instability may explain why persistence remains high while other features such as asymmetry appear less pronounced.

Given that Bitcoin is getting more integrated with the other financial markets including equities, commodities and foreign exchanges, these linkages, as Babalos et al. (2025) highlight, can amplify volatility through spillover effects and this growing integration of Bitcoin into the global financial markets may also introduce additional channels through which shocks can propagate and thus complicate the task of modeling volatility within a single-asset framework as is the case in this study.

Chapter 6: Conclusion

6.1 Summary of main findings

6.1.1 Volatility characteristics

The empirical analysis confirms that Bitcoin volatility exhibits the core stylised facts commonly observed in financial time series, including volatility clustering, strong persistence, and heavy-tailed behaviour, and the periods of elevated volatility tend to persist over time, while extreme movements occur more frequently than would be predicted under a Gaussian framework. These features highlight the importance of models capable of capturing time-varying conditional variance in cryptocurrency markets.

6.1.2 Model performance

Across all specifications, the GARCH(1,1) model under Gaussian innovations provides the strongest out-of-sample forecasting performance. Although IGARCH models capture high persistence by construction, they do not offer a clear improvement and in several cases perform worse than the standard GARCH specification, suggesting that full persistence may be too restrictive for Bitcoin volatility dynamics. Similarly, while Student-t innovations improve in-sample fit by accommodating heavy tails, these gains do not translate into better forecasting accuracy as the Gaussian specifications consistently outperform their heavy-tailed counterparts in out-of-sample evaluation, indicating that increased distributional flexibility may lead to overfitting rather than improved predictive performance. The Random Forest models perform substantially worse than the econometric benchmarks, and in spite of their flexibility in capturing nonlinear relationships, they fail to adequately model volatility persistence, resulting in higher forecast errors across all evaluation metrics.

6.1.3 Role of sentiment

The inclusion of search-based attention, proxied by Google Trends, does not materially improve forecasting performance in either the econometric or machine learning frameworks as the sentiment coefficient in GARCH-X and IGARCH-X models is statistically insignificant, while the Random Forest results show only negligible changes when sentiment variables are included. Therefore, the study concludes that Google Trends data is too general to rely on as the sole proxy for sentiment to be used for any meaningful inferences.

6.1.4 Robustness of results

The main findings remain stable across a range of robustness checks, including alternative volatility proxies, different evaluation windows, and variations in sentiment lag structures. The relative ranking of models does not change materially, with the GARCH(1,1) specification consistently outperforming both IGARCH and Random Forest models, thus, this stability indicates that the results are not driven by specific modeling choices or sample periods.

6.2 Contributions of the study

This study provides updated evidence on Bitcoin volatility dynamics using a long sample period spanning 2017 to 2025, a period that captures the major structural developments in the cryptocurrency markets, particularly with respect to Bitcoin, and the results demonstrate that standard GARCH models remain highly effective in forecasting volatility, even in a market characterised by rapid evolution and increasing institutional participation.

Methodologically, the study contributes by providing a direct and systematic comparison between GARCH-family models and Random Forests within a unified forecasting framework as well as by employing a rolling-window design and evaluating performance using QLIKE loss alongside SPA and MCS procedures, thus, ensuring that the analysis is robust and statistically rigorous comparison across model classes.

By highlighting the limitations of Google Trends as a representative of the search-based

attention measures, the findings contribute to the growing literature on behavioural finance and cryptocurrency markets that more elaborate and specific sentiment measures are needed to accurately capture the investor attention. While prior studies often report predictive gains from sentiment variables, this study shows that such gains are not universal and depend critically on the choice of proxy. In particular, aggregate search intensity may be too coarse to capture the behavioural mechanisms relevant for volatility formation.

6.3 Practical implications

The results suggest that relatively simple econometric models remain effective tools for volatility forecasting in cryptocurrency markets. For practitioners, this implies that well-specified GARCH models can provide reliable inputs for risk management, portfolio allocation, and Value-at-Risk calculations without requiring complex machine learning frameworks.

The findings also highlight the importance of aligning model choice with the underlying data-generating process and in the environments characterised by strong persistence and volatility clustering as is the case within the crypto market, the models that explicitly incorporate conditional variance dynamics outperform more flexible but structure-free alternatives, suggesting that model structure may be more important than flexibility in volatility forecasting tasks.

The weak performance of Google Trends indicates that not all sentiment proxies are equally informative and for the practitioners seeking to incorporate behavioural indicators, the results emphasise the need for more refined sentiment measures, such as text-based indicators derived from news or social media, rather than relying solely on aggregate attention metrics.

6.4 Future research directions

For improved sentiment measures, the study proposes that future research could explore more granular sentiment indicators derived from textual data sources such as news articles,

social media platforms, and online forums. These measures may better capture the direction and intensity of investor sentiment compared to aggregate search-based proxies.

Owing to the the relatively weak performance of standalone Random Forest models in this study, it is suggested that machine learning may be more effective when combined with econometric structures. Therefore, the hybrid models that integrate GARCH dynamics with nonlinear learning algorithms, such as LSTM or other deep learning architectures, represent a promising avenue for improving volatility forecasts in future studies.

The use of squared returns as a volatility proxy introduces measurement noise, and to mitigate this, future studies could employ high-frequency data to construct realised volatility measures, which may provide more accurate representations of the underlying intra-day volatility process and improve forecast evaluation. This should also be compared against other readily available volatility measures to determine how effective realized volatility is in comparison to the other measures.

Finally, given the evidence of structural changes and evolving market conditions, future research could extend the analysis to nonlinear econometric models, including regime-switching GARCH, smooth-transition models, and component GARCH specifications, models which may better capture the time-varying nature of cryptocurrency volatility.

References

Amoanu, S. (2025). Comparative forecasting of financial time series using arima, garch, random forest, and xgboost models. *Research Square*. Retrieved from https://doi.org/10.21203/rs.3.rs-7837766/v1 doi: 10.21203/rs.3.rs-7837766/v1

Ardia, D., Bluteau, K., & Rüede, M. (2019). Regime changes in bitcoin garch volatility dynamics. *Finance Research Letters*, *29*, 266-271. Retrieved from https://www.sciencedirect.com/science/article/pii/S1544612318303970 doi: https://doi.org/10.1016/j.frl.2018.08.009

Argente, D., & Van Patten, D. (2024, January 29). *El salvador adopted bitcoin as an official currency; salvadorans mostly shrugged.* Retrieved from https://insights.som.yale.edu/insights/el-salvador-adopted-bitcoin-as-an-official-currency-salvadorans-mostly-shrugged

Babalos, V., Bouri, E., & Gupta, R. (2025). Does the introduction of us spot bitcoin etfs affect spot returns and volatility of major cryptocurrencies? *The Quarterly Review of Economics and Finance*, *102*, 102006. doi: 10.1016/j.qref.2025.102006

Basher, S. A., & Sadorsky, P. (2022). Forecasting bitcoin price direction with random forests: How important are interest rates, inflation, and market volatility? *Machine Learning with Applications*, *9*, 100355. Retrieved from https://www.sciencedirect.com/science/article/pii/S266682702200055X doi: https://doi.org/10.1016/j.mlwa.2022.100355

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, *31*(3), 307–327.

Bollerslev, T., Engle, R. F., & Nelson, D. B. (1994). Arch models. *Handbook of Econometrics*, *4*, 2959–3038.

Cheikh, N. B., Zaied, Y. B., & Chevallier, J. (2020). Asymmetric volatility in cryptocurrency markets: New evidence from smooth transition garch models. *Finance Research Letters*,

35, 101293. Retrieved from https://www.sciencedirect.com/science/article/pii/S154461231930162X doi: https://doi.org/10.1016/j.frl.2019.09.008

Chu, J., Chan, S., Nadarajah, S., & Osterrieder, J. (2017). Garch modelling of cryptocurrencies. *Journal of Risk and Financial Management*, *10*(4). Retrieved from https://www.mdpi.com/1911-8074/10/4/17

Corsi, F. (2009, 02). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, *7*(2), 174-196. Retrieved from https://doi.org/10.1093/jjfinec/nbp001

Coulter, S. (2022). The impact of news media on bitcoin prices: Modeling data-driven discourses in the crypto-economy. *Digital Finance*, *4*(1), 25–54.

Demirer, R., Gkillas, K., Gupta, R., & Pierdzioch, C. (2022). Risk aversion and the predictability of crude oil market volatility: A forecasting experiment with random forests. *Journal of the Operational Research Society*, *73*(8), 1755–1767. Retrieved from https://doi.org/10.1080/01605682.2021.1936668 doi: 10.1080/01605682.2021.1936668

Diane, T., & Brijlal, P. (2024). Forecasting stock market realised volatility using random forest and artificial neural networks in south africa. *South African Journal of Economic and Management Sciences*, *27*(1), 1–15. Retrieved from https://doi.org/10.32479/ijefi.15431

Dudek, G., Fiszeder, P., Kobus, P., & Orzeszko, W. (2024). Forecasting cryptocurrencies volatility using statistical and machine learning methods: A comparative study. *Applied Soft Computing*, *151*, 111132. Retrieved from https://www.sciencedirect.com/science/article/pii/S156849462301150X doi: https://doi.org/10.1016/j.asoc.2023.111132

Dyhrberg, A. H. (2016). Bitcoin, gold and the dollar – a garch volatility analysis. *Finance Research Letters*, *16*, 85-92. Retrieved from https://www.sciencedirect.com/science/article/pii/S1544612315001038 doi: https://doi.org/10.1016/j.frl.2015.10.008

Engle, R. F. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, *50*(4), 987–1007.

Engle, R. F., & Bollerslev, T. (1986). Modelling the persistence of conditional variances. *Econometric Reviews*, *5*(1), 1–50. Retrieved from https://doi.org/10.1080/07474938608800095 doi: 10.1080/07474938608800095

Gensler, G. (2024, Jan 10). *Statement on the approval of spot bitcoin exchange-traded products.* Retrieved from https://www.sec.gov/newsroom/speeches-statements/gensler-statement-spot-bitcoin-011023

Han, B., Liu, A., Chen, J., & Knottenbelt, W. (2025). Can machine learning models better volatility forecasting? a combined method. *The European Journal of Finance*, *0*(0), 1–22. Retrieved from https://doi.org/10.1080/1351847X.2025.2553053 doi: 10.1080/1351847X.2025.2553053

Hansen, B. E. (1994). Autoregressive conditional density estimation. *International Economic Review*, *35*(3), 705–730. Retrieved from https://users.ssc.wisc.edu/~behansen/papers/ier_94.pdf

Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, *23*(4), 365–380. Retrieved from https://doi.org/10.1198/073500105000000063

Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, *79*(2), 453–497. Retrieved from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=522382

Hoang, L., & Vo, D. H. (2024). Google search and cross-section of cryptocurrency returns and trading activities. *Journal of Behavioral and Experimental Finance*, *44*, 100991. Retrieved from https://www.sciencedirect.com/science/article/pii/S2214635024001060 doi: https://doi.org/10.1016/j.jbef.2024.100991

Kabo, I. G., Obunadike, G. N., & Samaila, N. A. (2025). Sentiment-driven and economic indicators for bitcoin price forecasting: A hybrid time series model. *Journal of Science Research and Reviews*, *2*(1), 99–107. doi: 10.70882/josrar.2025.v2i1.34

Katsiampa, P. (2017). Volatility estimation for bitcoin: A comparison of garch models. *Economics Letters*, *158*, 3-6. Retrieved from https://www.sciencedirect.com/science/article/pii/S0165176517302501 doi: https://doi.org/10.1016/j.econlet.2017.06.023

Kim, H. Y., & Won, C. H. (2018). Forecasting the volatility of stock price index: A hybrid model integrating lstm with multiple garch-type models. *Expert Systems with Applications*, *103*, 25-37. Retrieved from https://www.sciencedirect.com/science/article/pii/S0957417418301416 doi: https://doi.org/10.1016/j.eswa.2018.03.002

Leber, T., & Egyed, E. (2025). The sentiment-augmented garch-lstm hybrid model for value-at-risk forecasting. *European Journal of Operational Research*. Retrieved from https://doi.org/10.1007/s10614-025-11042-8

Luong, C., & Dokuchaev, N. (2018). Forecasting of realised volatility with the random forests algorithm. *Journal of Risk and Financial Management*, *11*(4). Retrieved from https://www.mdpi.com/1911-8074/11/4/61

Mai, F., Shan, Z., Bai, Q., Wang, X. S., & Chiang, R. H. (2018). How does social media impact bitcoin value? a test of the silent majority hypothesis. *Journal of Management Information Systems*, *35*(1), 19–52. Retrieved from https://doi.org/10.1080/07421222.2018.1440774 doi: 10.1080/07421222.2018.1440774

Nelson, D. B. (1990). Arch models as diffusion approximations. *Journal of Econometrics*, *45*(1), 7-38. Retrieved from https://www.sciencedirect.com/science/article/pii/0304407690900928 doi: https://doi.org/10.1016/0304-4076(90)90092-8

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Business & Economic Statistics*, *29*(2), 183–204. Retrieved from https://doi.org/10.1016/j.jeconom.2010.03.034

Patton, A. J., & Sheppard, K. (2009). Evaluating volatility and correlation forecasts. *Handbook of Financial Time Series*, 801–838. Retrieved from https://ora.ox.ac.uk/objects/uuid:474e796d-5656-4e6b-94d3-b10125785fc5/files/mba0ba07178c0b21e3528d52e99430bd0

Sapkota, N. (2022). News-based sentiment and bitcoin volatility. *International Review of Financial Analysis*, *82*, 102183. Retrieved from https://www.sciencedirect.com/science/article/pii/S1057521922001454 doi: https://doi.org/10.1016/j.irfa.2022.102183

Wang, Y., Andreeva, G., & Martin-Barragan, B. (2023). Machine learning approaches to forecasting cryptocurrency volatility: Considering internal and external determinants. *International Review of Financial Analysis*, *90*, 102914. Retrieved from https://www.sciencedirect.com/science/article/pii/S1057521923004301 doi: https://doi.org/10.1016/j.irfa.2023.102914

Zahid, M., Iqbal, F., & Koutmos, D. (2022). Forecasting bitcoin volatility using hybrid garch models with machine learning. *Risks*, *10*(12). Retrieved from https://www.mdpi.com/2227-9091/10/12/237

Chapter A: Declaration of the Use of AI Tools

This thesis was prepared with limited assistance from generative artificial intelligence (AI) tools.

- CurreChat was used at an early stage to support the development of the overall structure and outline of the thesis.
- ChatGPT was used for limited purposes, including summarising selected academic literature, refining language, and improving clarity and structure during the drafting process.
- Grok was used to review and improve the language, coherence, and logical flow of the document.

All substantive components of the thesis, including the formulation of the research questions, methodological design, data collection, model implementation, empirical analysis, and interpretation of results, were carried out independently by the author. No AI-generated data, models, or empirical results have been used in this study.

As the author, I assume the full responsibility for the accuracy of the content and originality of this thesis work.